FINLITE: A WEB-BASED FINANCIAL LEDGER AND AUTOMATED REPORT GENERATION SYSTEM WITH AI CO-PILOT FOR THE LEAGUE OF INFORMATION TECHNOLOGY ENTHUSIASTS

A Project
Presented to the Faculty of the
College of Computer Studies
Pambayang Dalubhasaan ng Marilao

In Partial Fulfillment
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System Analysis and Design

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1. INTRODUCTION

This chapter presents the first part of the study, introducing the project context by outlining the proposed system, the target organization's background, the business logic and operational processes of the current system, and the specific problems encountered in existing manual operations.

1.1. Project Context

In contemporary organizational management, financial accountability hinges on the systematic recording of monetary flows and the punctual delivery of audit-compliant liquidation reports. A financial ledger serves as a centralized digital mechanism that systematically logs, categorizes, and monitors financial inflows and disbursements in real time, establishing operational transparency, strict custodial accountability, and verifiable audit trails. Complementing this is an automatic report generation system, an automated data processing engine that programmatically compiles, cross-verifies, and formats transactional data into standardized, audit-ready institutional reports, completely removing the vulnerabilities of manual calculation, arithmetic errors, and formatting non-compliance. FinLITE: A Web-Based Financial Ledger and Automatic Report Generation System with AI Chatbot integrates these two core functionalities into an operational financial co-pilot tailored to the unique administrative dynamics of the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng Marilao (PDM).

Pambayang Dalubhasaan ng Marilao (PDM) is a local tertiary institution established pursuant to municipal ordinances to provide accessible, high-quality higher education to the youth of Marilao and surrounding Bulacan municipalities, operating in compliance with the regulatory standards of the Commission on Higher Education (CHED) and the Association of Local Colleges and Universities (ALCU). Within PDM, the College of Computer Studies (CCS) is the primary computing division, offering the premier Bachelor of Science in Information Technology (BSIT) program under the executive academic leadership of Dean Dr. Emraida Marie M. Manucom and BSIT Program Director Jovylyn Ortiz-Cesar, MBA, MSIT.

The League of Information Technology Enthusiasts (LITE) is the officially accredited academic mother organization representing all students enrolled in the BSIT program at PDM. Guided by Club Advisers Ms. Kimberly Dawn Jatulan and Ms. Krizia Mae Genovia, LITE hosts programming contests, e-sports competitions, community outreach projects, and coordinates official representation in institutional functions like Institutional Club Week. LITE is governed by a multi-tiered leadership structure comprising an Executive Board—composed of the President, Vice President, Secretary, Treasurer, and Auditor—elected Year-Level Representatives for the 1st, 2nd, 3rd, and 4th Year cohorts, and nine specialized sub-committees covering Arduino, Cybersecurity, Programming, Networking, Multimedia, Photography, E-sports, Singing, and Dancing disciplines.

LITE operates under severe collegiate physical constraints, possessing no dedicated physical office, private conference room, or institutional workstation. Administrative planning, audit sessions, and ledger consultations take place transiently in campus corridors, student lounges, event booths, or at a communal faculty table located in the College of Computer Studies Office on the second floor. Money custody follows strict institutional protocols where all physical cash holdings, petty cash reserves, and accumulated coin collections are secured inside a single small metal cashbox stored inside a locked drawer within the Program Director's Office. Because the small metal cashbox never leaves the PDM campus, off-campus purchasing expeditions for event supplies cannot utilize upfront cashbox money directly. Furthermore, leadership transitions occur annually under a strict face-value turnover protocol, where the physical cash in the small metal cashbox must match the exact numerical balance attested on the audited final financial report before outgoing officers receive institutional clearance. Legitimate unpaid reimbursement claims remaining at the end of the academic year are certified and carried over as Accounts Payable liabilities to the incoming administration.

In accordance with established organizational practice, LITE does not collect mandatory student membership dues. Operating capital originates from four legitimate revenue streams classified systematically under organizational Inflows: student entry fees from competitive departmental events and tournaments; batch merchandise pre-order collections (such as departmental lanyards and organizational T-shirts), which strictly enforce a one-hundred percent upfront payment policy prior to vendor order placement, tracking uncollected student dues under Inflows until settled in full; booth rental fees (arkila) and sales commissions collected from concessionaires during Institutional Club Week; together with verified carried-over surplus reserves and certified opening liabilities inherited from prior academic tenures under the face-value turnover protocol.

Collection of organizational funds is conducted exclusively through physical cash transactions across collegiate activities. Physical cash collected at event booths by authorized executive officers is subject to a strict holding window, expected by the end of the day or the day after collection to be turned over to the Treasurer for deposit into the small metal cashbox in the Program Director's Office, though in practice this window is frequently delayed during intensive multi-day events. During active event booths, on-duty officers may execute emergency supply deductions directly from unremitted collections, provided an official commercial receipt or signed petty cash acknowledgment slip is submitted immediately upon turnover.

Disbursements for event supplies, certificates, sashes, trophies, and refreshments are governed by organizational authorization protocols, representing organizational Outflows. Organizational expenditures are disbursed directly from the physical cashbox; personal out-of-pocket advances, termed Abono, occur when there are insufficient funds or physical cash liquidity in the cashbox, or during off-campus purchasing errands. Under this practice, Club Advisers typically shoulder high-value expenses first, whereas student executive officers shoulder low-value supplies and refreshments out-of-pocket. These advances represent unliquidated organizational liabilities under Accounts Payable, liquidated through cashbox reimbursement payouts that are categorized strictly as Outflows, requiring rigorous tracking of both the original payer and the designated claimant requesting reimbursement.

Receipt submission is expected following activity completion, although strict enforcement regarding club advisers is often informal. Commercial purchases require official printed commercial receipts or sales invoices. Non-receipted small cash purchases, such as tricycle fares, public market craft purchases, or emergency printouts, require a handwritten expense acknowledgment slip detailing the date, monetary amount, travel route or item purpose, and claimant signature. If approved reimbursement claims exceed available physical cashbox liquidity, a partial cash payout is executed, disbursing available cash while logging the residual balance as an active pending liability under Accounts Payable.

Physical cash reconciliations are conducted periodically at the adviser's desk under a formal Dual-Officer Attestation Protocol. Operating the metal cashbox, the Treasurer acts as the Counter (physically sorting and counting banknotes and coins from one thousand pesos down to legal tender centavos into the interactive keypad), while the Auditor acts as the Verifier (witnessing the count and attesting the physical stacks against the ledger balance). The theoretical ledger balance is calculated by adding opening carryover balances and recorded inflows (including receivables), then subtracting reimbursed disbursements and operational outflows (including abono). The discrepancy variance is calculated by subtracting the theoretical balance from total counted physical cash. Balanced counts are certified at zero variance. If a shortage occurs, organizational spending and club activities continue as normal while the variance is investigated. Untraceable minor shortfalls may be declared as an allowable operating expense labeled Cash Shortage under written justification and dual adviser sign-off. Cash overages are retained inside the cashbox as unrecorded emergency petty cash buffers.

At semester end, LITE prepares a Final Financial Liquidation Report. Following digital draft review by Club Advisers, hard copies printed on institutional short bond paper are routed sequentially across a wet-ink signatory chain: starting with the Treasurer; followed by the Auditor; moving to the President; proceeding to both Club

Advisers for joint dual sign-off; moving to the BSIT Program Director for departmental review; and concluding with the Dean of the College of Computer Studies for final executive approval. If formatting or calculation errors are identified at later stages, the report is rejected, requiring manual re-editing, re-printing, and restarting the physical six-stage routing chain from the beginning.

Insights gathered through primary field interviews with the incumbent LITE President who served as treasurer and auditor across consecutive terms from 2024 to 2026, corroborated by club adviser questionnaires, reveal five critical operational problems within LITE's current manual financial workflows.

The first problem is personal financial vulnerability and delayed reimbursement of personal advances (*abono*). Because LITE cannot maintain an external operational bank account and must operate out of a single physical cashbox, off-campus purchasing runs require personal out-of-pocket advances whenever cashbox liquidity is insufficient. In such circumstances, Club Advisers are forced to shoulder high-value expenses (such as competition tokens, major event prizes, and large venue materials) first, while student executive officers shoulder low-value supplies and refreshments out-of-pocket. Thermal paper receipts gathered during fast-paced events frequently fade, crumple, or get lost before reaching the campus cashbox. Furthermore, delayed submission of receipts by student officers leaves reimbursement queues stalled for extended periods, clouding organizational liabilities. Loose receipt slips also lack two-person payment info clarifying who originally paid versus who is claiming reimbursement. When receipts are lost or submitted without proper authorization, expenditures cannot be audited, forcing advisers and officers to absorb these institutional costs personally.

The second problem is temporary documentation and administrative leakage in non-receipted small cash purchases. Non-receipted small cash purchases, such as jeepney and tricycle fares, local market craft purchases, and emergency photocopies, are currently documented through loose paper scraps, sticky notes, or unstructured Messenger chat screenshots. These informal records lack standardized information, including exact timestamps, itemized descriptions, authorized claimant identifiers, and supervising officer endorsements. Over a multi-month semester, these paper scraps are smudged or mislaid, depriving the Auditor of verifiable audit trails and resulting in unliquidated expense leakage.

The third problem encompasses accumulated cumulative ledger errors and forced personal out-of-pocket balance settlements during administrative turnovers. Because daily transaction tracking and paper ledger updates are frequently delayed until weeks or months after major departmental events, underlying bookkeeping errors, duplicate entries, unrecorded petty cash disbursements, and miscalculated cash collections accumulate unnoticed throughout the academic term. At the end of the officers' administrative tenure, when physical cash in the metal cashbox is cross-referenced against the theoretical ledger balance for the mandatory semester-end clearance audit, these accumulated disparities manifest as unexplained cash shortfalls. Under existing institutional turnover protocols, outgoing officers cannot obtain clearance for graduation, leadership turnover, or academic records until the physical cashbox balance perfectly aligns with the audited financial report. As a result, outgoing student officers and club advisers are compelled to cover these financial gaps out of their own pockets, absorbing the costs of unlogged institutional expenses simply to secure their required graduation clearance.

The fourth problem involves latent discrepancies and slow manual cash computations. Physical cash counts, sorting bills from one thousand pesos down to one-peso bills, are performed weeks or months after events conclude rather than on a regular cadence. In the absence of automated reconciliation tools, officers perform manual cross-multiplication against loose paper tallies, hiding arithmetic errors until final semester report compilation. When cash variances emerge, the resulting mandatory spending and activity freeze paralyze club operations while officers struggle to determine whether the disparity represents an unlogged petty disbursement, an unremitted collection deduction, or an untraceable coin variance.

The fifth problem consists of institutional liquidation constraints and signatory rejections. Manual preparation of end-of-term liquidation reports using basic word processors results in formatting non-compliance with College of Computer Studies standards, including uneven table borders, misaligned decimal figures, and inconsistent font sizes. Navigating the physical six-stage wet-ink signatory chain, including the mandatory dual adviser joint sign-off, becomes a major delay in the process. If a calculation error, missing receipt attachment, arithmetic error, or typographical defect is caught by the Program Director or Dean at the fifth or sixth stage, the report is rejected. Under manual operations, officers must manually re-compute, re-edit, reprint the multi-page document on institutional bond paper, and restart physical wet-ink routing from the first stage, causing immense administrative delay during final examination and graduation periods.

1.2. Objectives of the Study

1.2.1. General Objectives

The general objective of the study is to design and develop FinLITE: A Web-Based Financial Ledger and Automatic Report Generation System with an AI Chatbot for the League of Information Technology Enthusiasts (LITE) at Pambayang

Dalubhasaan ng Marilao to streamline organizational financial transaction tracking, automate institutional liquidation report compilation, and provide intelligent audit assistance.

1.2.2. Specific Objectives

This project has the following specific objectives:

1. To design and implement a secure web-based financial ledger with configurable role-based access control and dynamic administrative permission checkboxes that records cash collections and student installment receivables (hulog) as Inflows, manages personal out-of-pocket advances (abono) as Outflows, and establishes a two-stage transaction audit verification and email revision feedback loop to ensure financial accountability and prevent unliquidated expense leakages.
2. To develop an interactive cash denomination counter and automated variance reconciliation engine that tallies physical bills and coins against single-entry ledger balances down to centavos under dual-officer attestation (Treasurer as Counter and Auditor as Verifier) to eliminate arithmetic errors, record cash shortage justifications, and monitor custodial turnovers.
3. To deploy a one-click institutional document generation engine that compiles CCS-compliant liquidation reports in editable Word (.docx) and PDF formats, integrated with an AI Chatbot co-pilot accessible across all officer roles (Admin, President, Vice President, Treasurer, Auditor) to provide instant budget and expense inquiries.

1.3. Purpose and Description

This study is significant to the following:

1.3.1 LITE Organization Student Officials

The system benefits student officials by replacing paper records with an auditable digital ledger, classifying student receivables under inflows and personal advances (*abono*) under outflows, ensuring prompt reimbursement of advances, empowering the President and Vice President to resolve audit discrepancies through role-gated transaction editing, protecting the Treasurer from tampering allegations by restricting unilateral edits on flagged records, and safeguarding outgoing officers from paying cash shortages out-of-pocket during turnover.

1.3.2 Club Advisers

The system benefits Club Advisers by granting them overall management and real-time visibility over organizational funds, empowering them with a dynamic permission matrix to configure and delegate officer access via checkboxes at any time, streamlining the digital review and approval of reimbursement claims, and allowing them to query the AI Chatbot for instant budget updates.

1.3.3 LITE Auditor

The system benefits the Auditor by establishing a dedicated transactional gatekeeping interface where all inflows and outflows are verified against supporting receipts and disbursement records, providing automated tools to approve records or dispatch revision directive emails to specified or all officers, and ensuring that flagged transactions remain visible on the active dashboard to prevent ledger distortion.

1.3.4 BSIT Students

The system will benefit the general student body of the Bachelor of Science in Information Technology program by upholding financial integrity and accountability in the management of student activity fees, merchandise payments, and event booth revenues, thereby ensuring that organizational funds are utilized efficiently for meaningful collegiate technical activities and student development.

1.3.5 BSIT College Leadership

The system benefits the Program Director and the Dean by ensuring all submitted financial reports follow correct CCS formatting and math, avoiding repeated document rejections and delays.

1.3.6 The Proponents

The system benefits the researchers by helping them apply their skills in web programming, database design, and AI integration to solve a real campus problem.

1.3.7 Future Researchers

The system benefits future researchers by serving as a useful guide and reference for building financial tools for student organizations.

1.4. Scope and Delimitation

This study focuses on the design and development of FinLITE, a web-based financial ledger and automated report generation system with an AI Co-Pilot for the League of Information Technology Enthusiasts (LITE) at Pambayang Dalubhasaan ng

Marilao (PDM). The system starts with user authentication and is divided into specific modules aimed at replacing the organization's manual record-keeping and solving common financial tracking issues.

The system workflow initiates with the User Authentication and Dynamic Role-Based Access Control Module, where the LITE president and vice president, treasurer, auditor, and club advisers securely register and log in to the web platform using validated @gmail.com accounts via Google OAuth 2.0 or email and password. Upon credential verification and approval at the administrative waiting gate, the Club Advisers assume overall system management as System Administrators. Rather than enforcing rigid, unchangeable access silos, the platform provides an interactive permission delegation matrix in the Admin Console (/admin) using configurable checkboxes. Club Advisers can inspect any officer's account and dynamically grant, customize, or disable operational capabilities at will—such as permitting the Auditor to encode inflows or outflows during emergency event drives that normally only the Admin, President, Vice President, or Treasurer are authorized to handle, or vice versa.

By default, the system enforces institutional segregation of duties: the Treasurer is authorized to query the AI Co-Pilot and encode cash collections, tournament fees, and installment receivables (hulog) under Inflows, as well as record operational disbursements and abono reimbursement payouts under Outflows, operate the digital denomination keypad during physical cash reconciliations, and compile liquidation reports. The Auditor is authorized to query the AI Co-Pilot and audit all transactions, serving as the verification checkpoint through which all inflows and outflows must pass. When auditing, the Auditor inspects receipts and figures: if correct, the Auditor approves the entry, updating the database and dashboard to a VERIFIED status; if details are incorrect, the transaction still reflects on the active database and dashboard to preserve cashbox transparency, while the Auditor dispatches an automated email containing their feedback and transaction details to specified officers or broadcasts to all for revision. The LITE President and Vice President are authorized to query the AI Co-Pilot, record inflows and outflows, and exercise exclusive transaction editing authority: upon receiving the Auditor's revision email, they follow a secure deep-redirect link into FinLITE, amend the transaction details and receipts, and resubmit it back to the Auditor for re-checking. Conversely, if the Treasurer receives a revision email, they are redirected to view the full details and auditor remarks but are strictly locked from editing to prevent unilateral tampering by the cash custodian, unless the Club Adviser explicitly checks the edit permission box for their account. The Club Advisers maintain overall administrative authority, real-time dashboard oversight, dual sign-off on finalized reports, and AI CoPilot query access.

Once logged in, the Club Adviser and executive officers access the Advance & Reimbursement Management Module to resolve personal financial vulnerability and delayed reimbursement of out-of-pocket advances (*abono*), classified under organizational Outflows. In accordance with organizational rules, purchases are primarily funded directly from the physical cashbox; personal advances only occur when there are insufficient funds or physical liquidity in the cashbox. When funds are depleted, Club Advisers shoulder high-value expenses (such as competition tokens, major event prizes, and large venue materials) first, while student executive officers shoulder low-value supplies and refreshments out-of-pocket. When an officer or Club Adviser advances personal funds, they open this module on their smartphone or computer to file an immediate claim. Using a dual-entity tracking form, the claimant designates both the original payer and the designated claimant, then uploads a photo of the commercial sales invoice. Through a restricted interface, the Auditor performs real-time receipt tracking and verification to inspect uploaded proofs and catch missing claimant details early, while the Club Advisers execute digital approval of the *abono* claim. Meanwhile, the LITE President or Vice President monitors outstanding reimbursement debt on the budget dashboard, and the Treasurer accesses the *abono* payout queue to execute cashbox reimbursements, systematically logging full or partial payout states and updating creditor balances to ensure no leader absorbs organizational expenses out of pocket.

To address the second problem of temporary documentation and administrative leakage in non-receipted micro-disbursements, the authorized purchasing officer—such as the LITE President, Vice President, or Treasurer—accesses the Non-Receipted Expense & Transit Logging Interface embedded within the unified Outflows recording module. For low-value transit and incidental operational expenses that inherently lack official commercial receipts, such as tricycle and jeepney fares between PDM and Marilao Poblacion, emergency photocopies, or public market craft purchases, the purchasing officer logs the disbursement directly on screen using standardized travel route presets or operational expense categories. The officer inputs the itemized description, date, and monetary amount, capturing digital claimant verification alongside the supervising endorsement of the LITE President or Vice President. Once submitted, the system records the expense directly under Outflows with full audit metadata; the Auditor accesses the verification interface to inspect the logged particulars, appending audit remarks and tracking claimant accountability in real time, while the Treasurer records the corresponding cashbox disbursement, permanently eliminating fragile paper scraps, sticky notes, and unstructured chat screenshots.

The Treasurer and LITE President or Vice President then transition to the Witnessed Custodial Tracking and Turnover Module to resolve physical cashbox custodial risks and high-stress face-value turnover obligations. Upon remittance, the Treasurer encodes the turned-over cash into the single-entry ledger under Inflows. At the conclusion of an academic tenure, the outgoing Treasurer and LITE President and Vice President run the turnover protocol witnessed by the Auditor and Club Advisers, generating an official Turnover Reconciliation Certificate that validates counted physical cash against audited ledger balances down to the exact peso. The module formally archives legitimate unsettled reimbursement claims as certified Accounts Payable carryover liabilities for the incoming administration, safeguarding outgoing leaders from unjustified out-of-pocket balance deductions during graduation clearance.

To eliminate latent arithmetic errors and prevent paralyzing club-wide spending freezes, the Treasurer and Auditor access the Interactive Cash Denomination Counter and Variance Reconciliation Engine during periodic physical audits conducted at the adviser's desk. Operating under the Dual-Officer Attestation Protocol, the Treasurer functions as the Counter, inputting the physical quantity of each bill (₱1,000, ₱500, ₱200, ₱100, ₱50, ₱20) and coin denomination (₱20, ₱10, ₱5, ₱1, and centavos) into the digital keypad calculator, while the Auditor functions as the Verifier, inspecting the physical count and co-signing the session. The engine instantly computes the total cash on hand and cross-references it in real time against the theoretical ledger balance, triggering an automated variance alert for the Auditor if a discrepancy is detected. Under the discrepancy logging protocol, if a verified untraceable minor shortfall occurs, the system automatically drafts a standardized Cash Shortage Justification Report for the Auditor to log and the Club Advisers to endorse through digital dual sign-off. Conversely, any cash overage is credited directly to the emergency petty cash buffer, maintaining transparent accountability while preventing unwarranted operational freezes.

Finally, the executive officers navigate to the one-click Institutional Document Generation Engine and AI Co-Pilot to solve institutional liquidation bottlenecks, formatting non-compliance, and signatory rejections. With a single click, the Treasurer triggers the export engine, which pulls all validated ledger entries, supporting receipt attachments, and reconciliation logs to programmatically compile end-of-term financial reports in Microsoft Word (.docx) and PDF formats. These reports strictly comply with College of Computer Studies (CCS) standards, featuring Times New Roman typography, 12pt font sizing, borderless tables, right-aligned two-decimal currency figures, and the pre-formatted wet-ink signatory block. Directly in-browser, the LITE President or Vice President previews the exact report layout to make live text and metadata adjustments before export, and possesses the capability to unlock returned reports for quick re-export without re-typing the entire document from scratch. The report then progresses through the system's digital routing workflow, securing online certification from the Auditor at Stage 2, executive approval from the LITE President or Vice President at Stage 3, and digital dual sign-off from both Club Advisers at Stage 4 prior to physical printing for the Program Director and Dean. Concurrently, an embedded zero-hallucination AI Co-Pilot is accessible across the platform to all four active roles (Club Advisers/Admin, President/VP, Treasurer, and Auditor), allowing officers to submit natural-language inquiries to check remaining event budgets, monitor reimbursement debts, and inspect category totals instantly without manual spreadsheet combing.

The operational boundaries of this study are defined by several delimitations and governance safeguards to preserve institutional compliance and feasibility. First, FinLITE operates strictly as a physical cash ledger, denomination counter, and audit tracking system, and does not integrate digital financial payment gateways, mobile wallet interfaces (such as GCash or Maya), electronic fund disbursements, or banking APIs; all recorded revenues originate from physical cash collections, booth receipts, and upfront merchandise payments, while student installment receivables (hulog) are credited to Inflows upon physical cash remittance. Second, the system does not encompass automated physical currency handling, mechanical coin sorting, or electronic locking mechanisms for the physical metal cashbox stored inside the Program Director's Office, operating strictly as a digital ledger, calculation, and audit tracker. Third, to preserve unbroken audit integrity across academic tenures, the platform enforces a Financial Audit Preservation Rule that prohibits the deletion of user accounts linked to existing transactions, cash reconciliations, or audit remarks, restricting administrative action strictly to account suspension. Fourth, API rate-limiting middleware restricts request volumes across authentication, report generation, and AI chat endpoints to protect server stability, while outbound email dispatches (such as registration alerts, password reset tokens, and auditor revision notices) are handled via secure SMTP delivery.

Fifth, while the platform automates report calculation, layout formatting, and pre-audit validation, it does not bypass or replace the physical wet-ink signatory routing chain required by collegiate governance; final generated reports must still be printed on institutional bond paper and manually signed across all administrative stages for official institutional filing. Lastly, the system’s workflow rules, approval protocols, and reporting templates are customized exclusively for the League of Information Technology Enthusiasts (LITE) under the College of Computer Studies at Pambayang Dalubhasaan ng Marilao, and do not apply to other academic organizations, campus student councils, or external academic departments.

2. TECHNICAL BACKGROUND OF THE STUDY

2.1 Project Analysis

The analysis phase thoroughly investigated the existing financial tracking, cash custody, and liquidation routines of LITE across successive administrations from 2024 to 2026. Data gathered through an in-depth qualitative interview with the 2026–2027 LITE President, who previously served as organizational Auditor and Treasurer—corroborated by structured Google Forms questionnaires administered to the Faculty Club Advisers, revealed critical inefficiencies in tracking receipts, the mixing of personal and organizational funds, manual cash calculation errors, and delays in paper-based report approvals. To establish a rigorous architectural foundation, the operational workflows of both the existing manual system and the proposed FinLITE platform were formally modeled using structured Data Flow Diagrams (DFDs) following standard Gane-Sarson notation.

2.1.1 Existing Data Flow Diagram

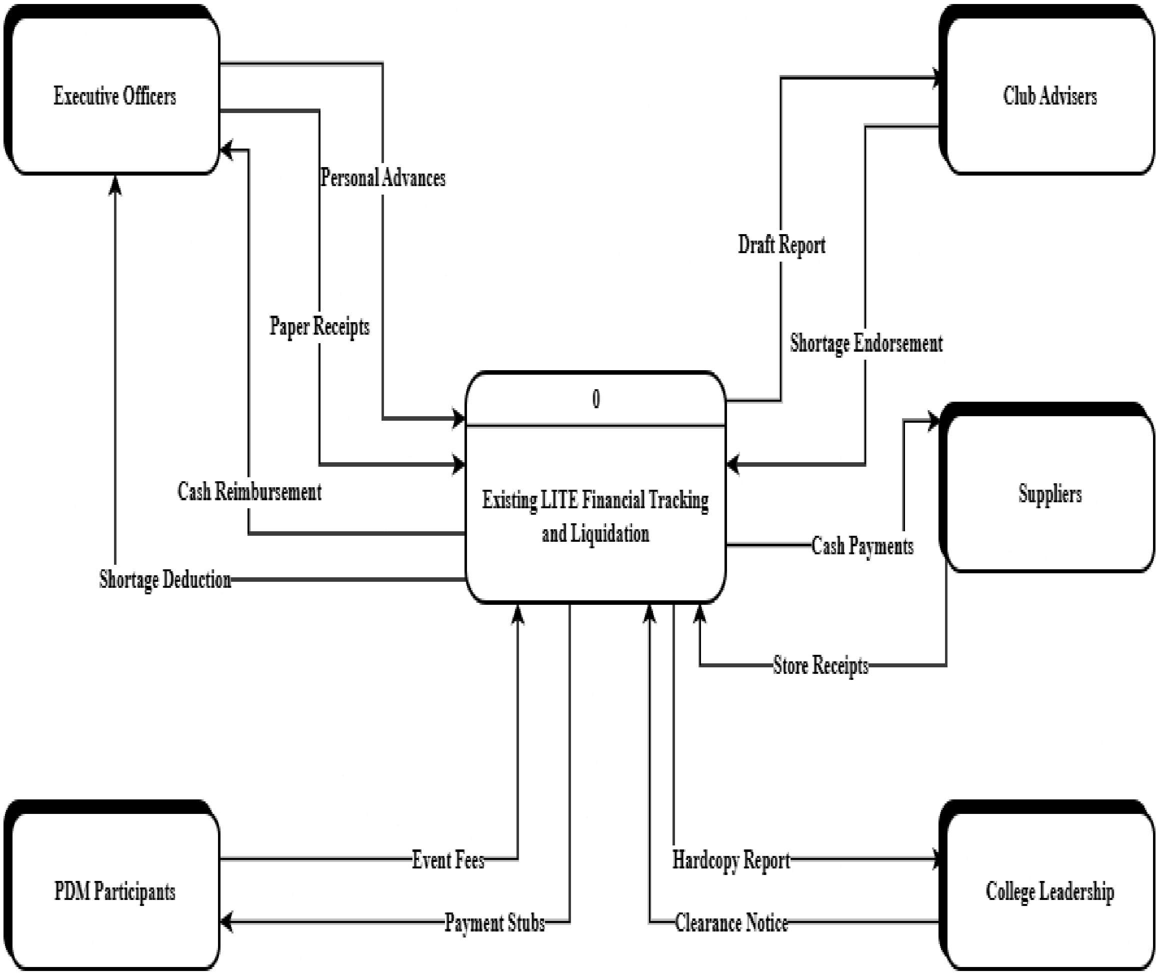

Executive Officers
Club Advisers
Personal Advances
Draft Report
Paper Receipts
Shortage Endorsement
0
Suppliers
Cash Reimbursement
Existing LITE Financial Tracking and Liquidation
Cash Payments
Shortage Deduction
Store Receipts
PDM Participants
Event Fees
Hardcopy Report
College Leadership
Payment Stubs
Clearance Notice

Figure 1: Context Diagram of the Existing LITE Financial Tracking and Liquidation

Figure 1 illustrates the Context Diagram of the Existing LITE Financial Tracking and Liquidation, defining the high-level system boundary and external information exchanges governing the student organization's current manual operational workflow. In this manual framework, financial operations are driven by physical currency exchanges, informal communication channels, and loose paper documentation. When student executive officers execute emergency purchasing runs for event supplies, certificates, trophies, and refreshments, they front personal capital (abono) and collect printed thermal paper receipts from external suppliers, which are subsequently surrendered to the Treasurer alongside reimbursement claims. Tournament participants and general students pay physical cash registration fees directly to on-duty officers, receiving handwritten paper acknowledgment stubs or event wristbands in return. In turn, commercial vendors and external suppliers receive direct cash payments from officers or the Treasurer upon issuing physical store receipts.

Physical cash holdings are held by student collectors during events and later deposited into a small metal cashbox. Prior to final submission, the Treasurer compiles a draft liquidation report for adviser inspection. At the conclusion of the academic semester, the Treasurer manually formats a multi-page hardcopy liquidation report using desktop word processors, which must navigate a physical, sequential wet-ink signatory chain (Treasurer, Auditor, President/Vice President, Dual Club Advisers, Program Director, and CCS Dean). College leadership reviews the completed dossier and issues an institutional clearance notice only after verifying all mathematical computations and supporting physical receipts, with any single discrepancy requiring document re-printing and restarting the wet-ink clearance cycle.

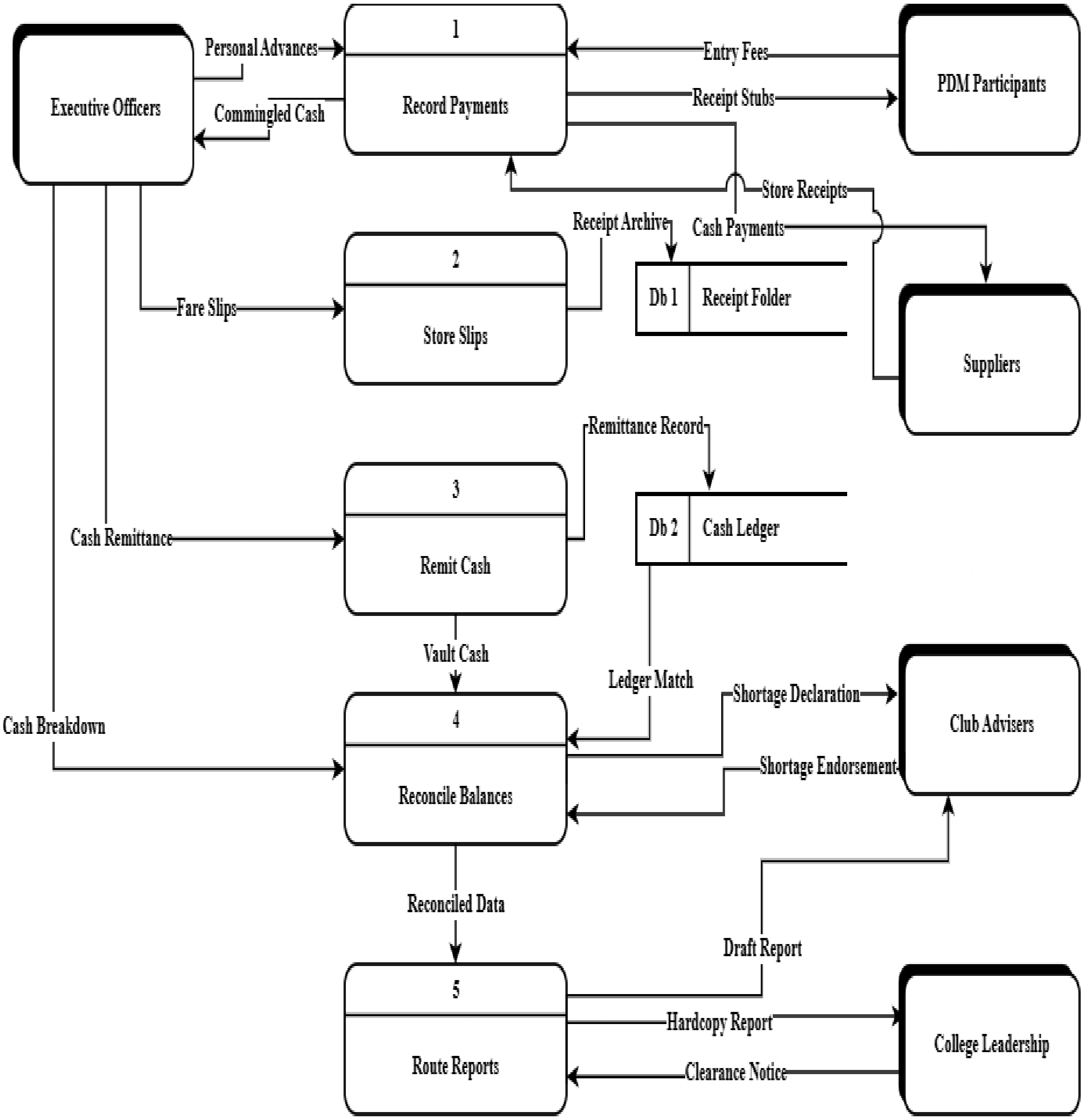

Executive Officers
Personal Advances
Commingled Cash
1
Record Payments
Entry Fees
Receipt Stubs
PDM Participants
Store Receipts
Cash Payments
Receipt Archive
2
Store Slips
Fare Slips
Db 1
Receipt Folder
Suppliers
Remittance Record
3
Remit Cash
Cash Remittance
Db 2
Cash Ledger
Vault Cash
Ledger Match
Cash Breakdown
4
Reconcile Balances
Shortage Declaration
Shortage Endorsement
Club Advisers
Reconciled Data
Draft Report
5
Route Reports
Hardcopy Report
Clearance Notice
College Leadership

Figure 2: Existing Diagram 0 of LITE Financial Tracking and Liquidation

As illustrated in Figure 2, executive officers draw from organizational funds when preparing for events, dipping into personal capital whenever cash reserves prove insufficient to purchase supplies from local vendors and merchants.. In return, the suppliers hand them printed thermal receipts, which the officers keep inside a paper folder. At the same time, students participating in the event pay their registration fees in cash and receive a simple tear-off paper stub as proof of payment. Because officers collect this cash on the go, the club money often mixes with their personal funds in their pockets and wallets, which they later use to pay themselves back for what they spent in advance. For travel expenses like jeepney and tricycle rides where drivers cannot issue official receipts, the officers write down their fare details on handwritten slips and store them in the folder alongside the other receipts.

Once collections and errands are done, the officers hand over the collected cash to the treasurer. The treasurer notes down who is holding the funds, writes the remittance into a physical ledger notebook, and places the physical bills and coins into storage. When it is time to check the funds, the treasurer pulls out the stored cash and manually counts every bill and coin. This physical count is compared with the previous balance recorded in the ledger to see if the amounts match. Should any financial discrepancies or shortages arise, officers must report them to their club advisers to provide clarification and seek necessary approval. Finally, the treasurer prepares a draft report and submits it to the club advisers for initial review, gathers all the paper receipts from the folder, matches them with the records in the ledger, and prints out a final hardcopy report to submit to college leadership for institutional review and clearance.

2.1.2 Proposed Data Flow Diagram

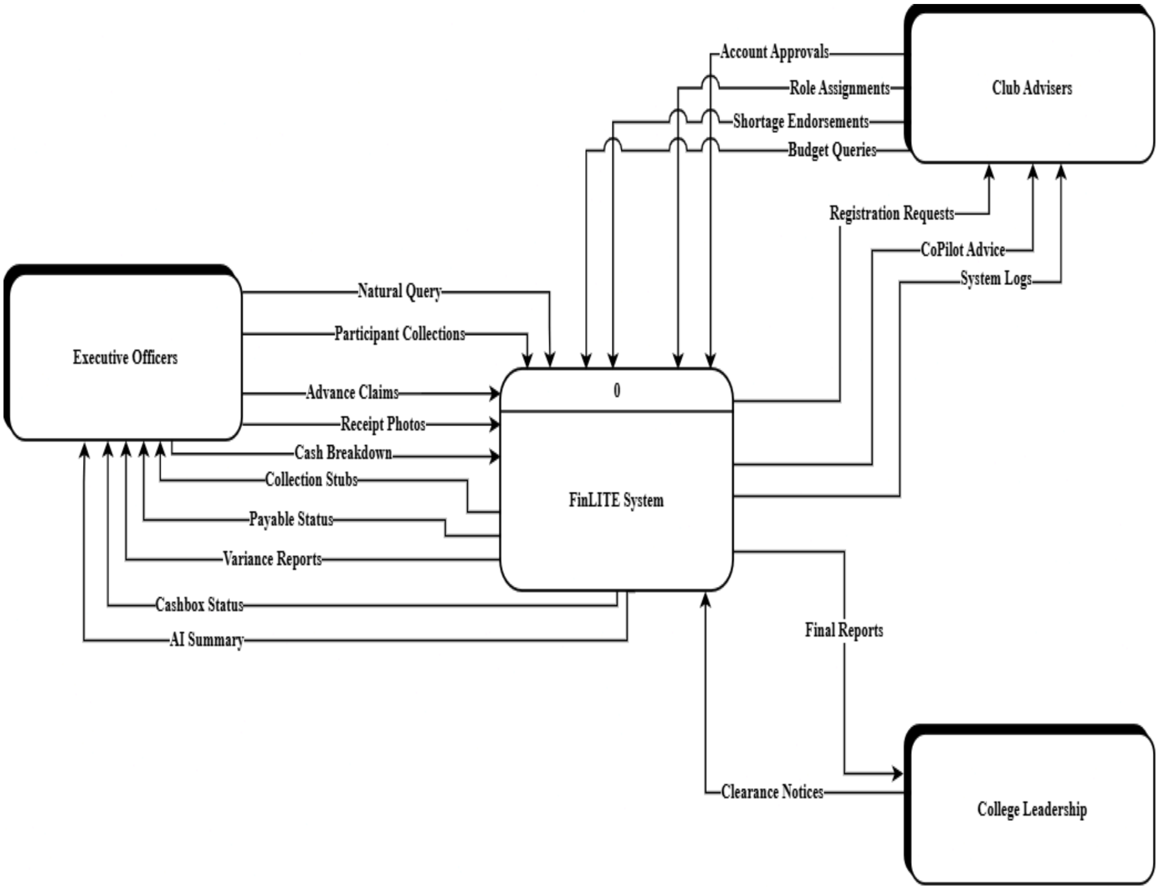

Account Approvals
Role Assignments
Shortage Endorsements
Budget Queries
Club Advisers
Registration Requests
CoPilot Advice
System Logs
Executive Officers
Natural Query
Participant Collections
Advance Claims
Receipt Photos
Cash Breakdown
Collection Stubs
Payable Status
Variance Reports
Cashbox Status
AI Summary
0
FinLITE System
Final Reports
Clearance Notices
College Leadership

Figure 3: Context Diagram of FinLITE System

Figure 3 illustrates the Context Diagram (Level 0) of the proposed FinLITE: A Web-Based Financial Ledger and Automated Report Generation System with AI Co-Pilot for the League of Information Technology Enthusiasts at Pambayang Dalubhasaan ng Marilao. FinLITE is represented as a singular top-level process, interfacing with three primary external institutional entities: Executive Officers, Club Advisers, and College Leadership.

Operational custody of the cashbox belongs to Executive Officers (President/VP, Treasurer, Auditor), who log inflows, advance claims, receipt photos, and cash breakdowns to receive collection stubs, payable status, variance reports, cashbox updates, and AI summaries.

Club Advisers manage governance and administration by handling account approvals, role assignments, shortage endorsements, and budget queries, while receiving registration requests, AI advice, and system logs.

College Leadership (BSIT Program Director and CCS Dean) serves as the clearance authority, reviewing compiled final reports to issue institutional clearance notices for organizational accreditation.

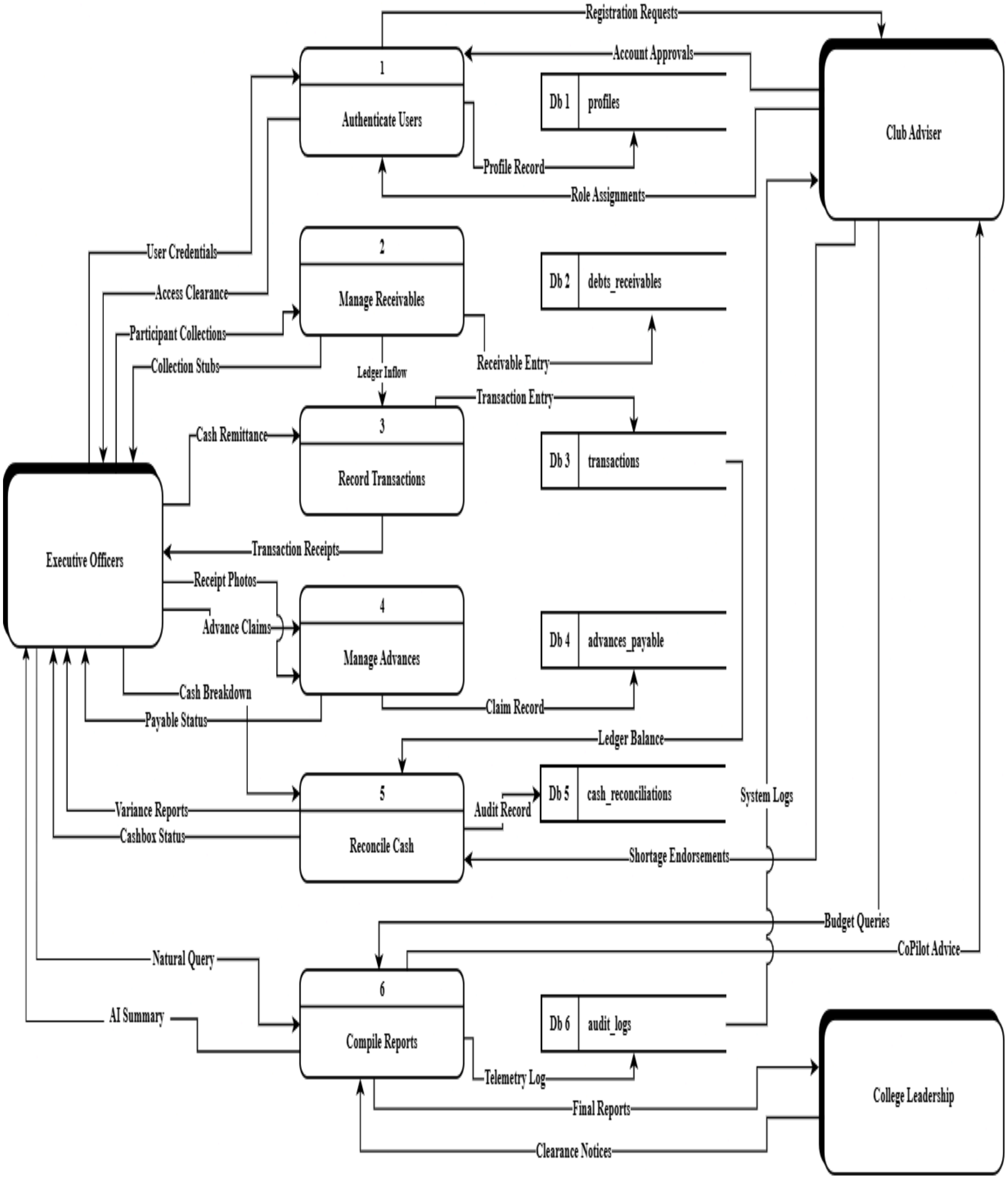

Registration Requests
Account Approvals
1
Authenticate Users
Db 1
profiles
Profile Record
Role Assignments
Club Adviser
User Credentials
Access Clearance
Participant Collections
Collection Stubs
2
Manage Receivables
Db 2
debts_receivables
Receivable Entry
Ledger Inflow
Transaction Entry
Cash Remittance
3
Record Transactions
Db 3
transactions
Executive Officers
Transaction Receipts
Receipt Photos
Advance Claims
4
Manage Advances
Db 4
advances_payable
Cash Breakdown
Payable Status
Claim Record
Ledger Balance
5
Reconcile Cash
Db 5
cash_reconciliations
Audit Record
System Logs
Variance Reports
Cashbox Status
Shortage Endorsements
Budget Queries
CoPilot Advice
Natural Query
AI Summary
6
Compile Reports
Db 6
audit_logs
Telemetry Log
Final Reports
Clearance Notices
College Leadership

Figure 4: Diagram 0 of FinLITE System

Figure 4 presents the proposed FinLITE platform, breaking down the main system into six functional processes supported by six relational databases.

In Process 1 (Authenticate Users), officers register and log in using Google OAuth 2.0 or standard credentials. Pending accounts wait in a restricted view until Club

Advisers approve them, assign operational roles in Db 1 (profiles), and grant access clearance. In Process 2 (Manage Debts), officers log merchandise pre-orders and installment collections into Db 2 (debts_receivables) and issue electronic stubs. Settled accounts automatically transfer as inflows to Process 3. In Process 3 (Record Transactions), the Treasurer logs income and expenditures with attached receipt photos into Db 3 (transactions) after validating available cashbox liquidity.

In Process 4 (Manage Advances), officers submit out-of-pocket advance claims and receipts into Db 4 (advances_payable). The Treasurer verifies receipts and issues full or partial reimbursements from the cashbox. In Process 5 (Reconcile Cash), the Treasurer inputs physical cash counts across 11 Philippine denominations into Db 5 (cash_reconciliations). The system calculates variances against ledger balances from Db 3, sending verified shortages to Club Advisers for endorsement. In Process 6 (Compile Reports), system logs are saved to Db 6 (audit_logs) while the engine aggregates data from Db 3, Db 4, and Db 5. Authorized users can query the AI Co-Pilot for budget insights or export final liquidation reports to College Leadership for institutional clearance.

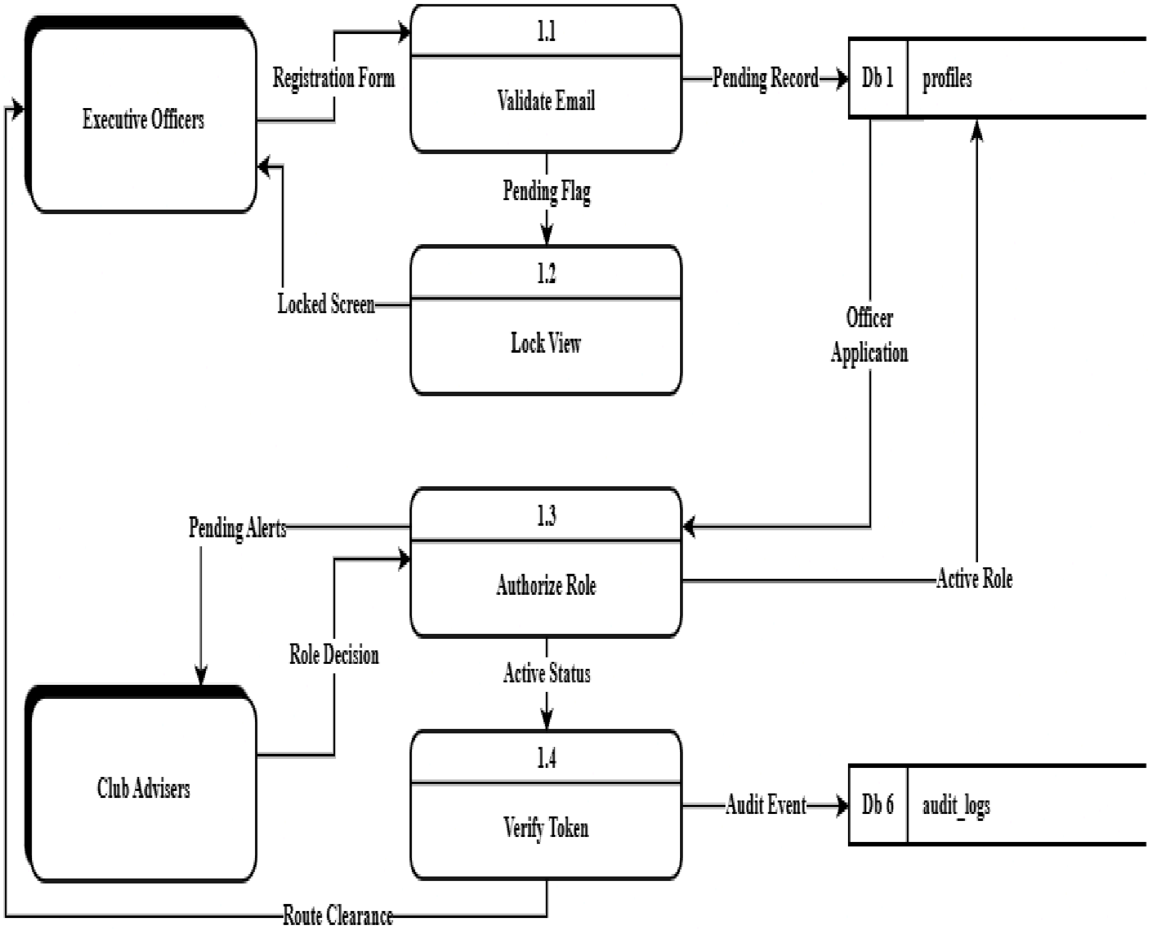

1.1
Validate Email
Executive Officers
Registration Form
Pending Record
Db 1
profiles
Pending Flag
1.2
Lock View
Locked Screen
Officer Application
1.3
Authorize Role
Pending Alerts
Role Decision
Active Role
Active Status
1.4
Verify Token
Club Advisers
Audit Event
Db 6
audit_logs
Route Clearance

Figure 5: Diagram 1 of Process 1: Authenticate Users

Figure 5 illustrates the Level 2 decomposition of Process 1.0 (Authenticate Users) across four key steps. When an applicant officer submits a Registration Form, Sub-process 1.1 (Validate Email) checks their credentials and creates a pending profile in Db 1 (profiles). Sub-process 1.2 (Lock View) then holds unapproved accounts in a quarantined state, showing a Locked Screen to the applicant while sending pending alerts to the Club Advisers. Following adviser review, Sub-process 1.3 (Authorize Role) receives the adviser's Role Decision and updates the user's account status to active. Finally, Sub-process 1.4 (Verify Token) issues session tokens to authenticated officers and logs the login event in Db 6 (audit_logs).

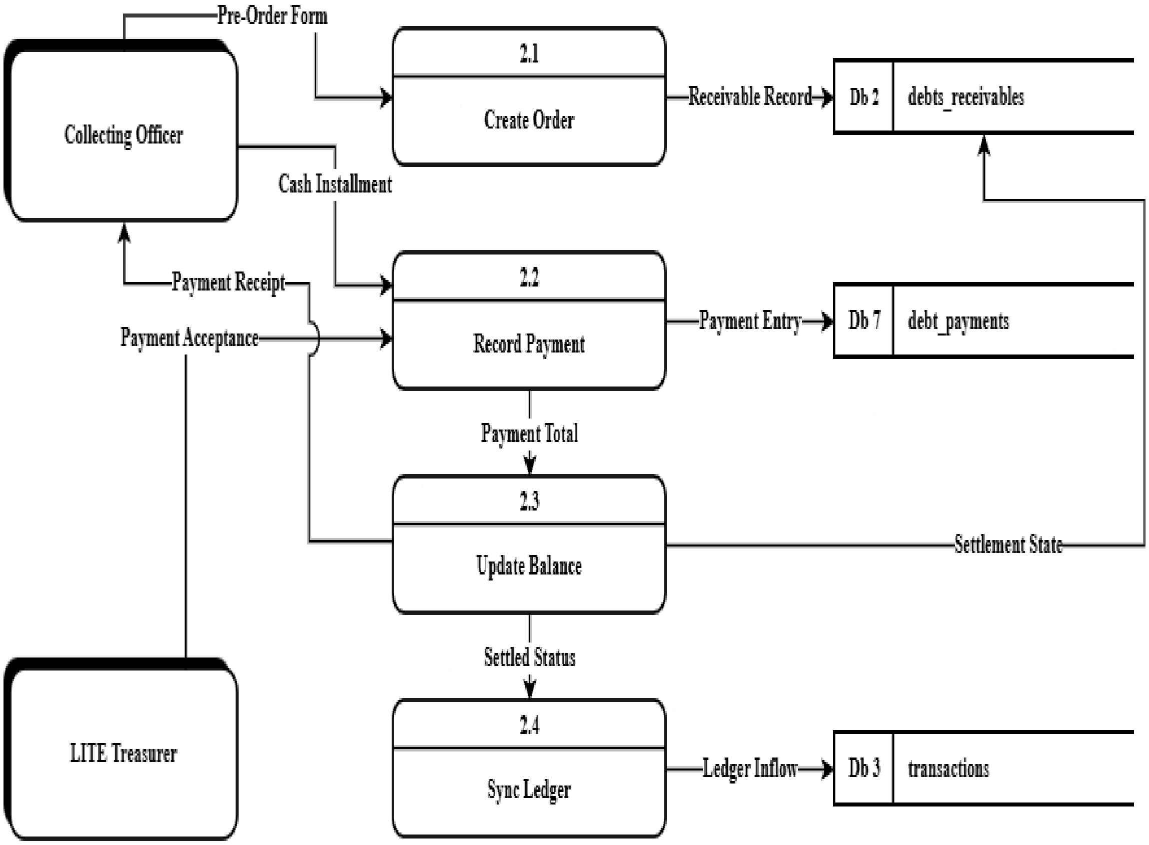

Pre-Order Form
Collecting Officer
2.1
Create Order
Receivable Record
Db 2
debts_receivables
Cash Installment
Payment Receipt
Payment Acceptance
2.2
Record Payment
Payment Entry
Db 7
debt_payments
Payment Total
2.3
Update Balance
Settlement State
Settled Status
LITE Treasurer
2.4
Sync Ledger
Ledger Inflow
Db 3
transactions

Figure 6: Diagram 1 of Process 2: Manage Receivables

Figure 6 outlines the Level 2 decomposition of Process 2 (Manage Receivables) across sub-processes that manage organizational receivables requested by individuals, such as faculty members.

In Sub-process 2.1 (Create Order), an authorized officer submits a request detailing the borrower’s name, contact details, and requested loan specifications, creating an active receivable profile in Db 2 (debts_receivables). Proceeding directly to Sub-process 2.2 (Record Payment), when the borrower repays cash, Sub-process 2.2 receives the cash repayment countersigned by the Treasurer's Payment Acceptance, recording each payment entry into Db 7 (debt_payments).

Sub-process 2.2 passes the Payment Total to Sub-process 2.3 (Update Balance), which calculates the total payments against the contract amount, updates the Settlement State in Db 2 (Unpaid, Partially Paid, Settled), and issues an electronic Payment Receipt. Once fully settled, Sub-process 2.3 triggers Sub-process 2.4 (Sync Ledger) to record a verified Ledger Inflow in Db 3 (transactions), keeping receivables aligned with cash holdings.

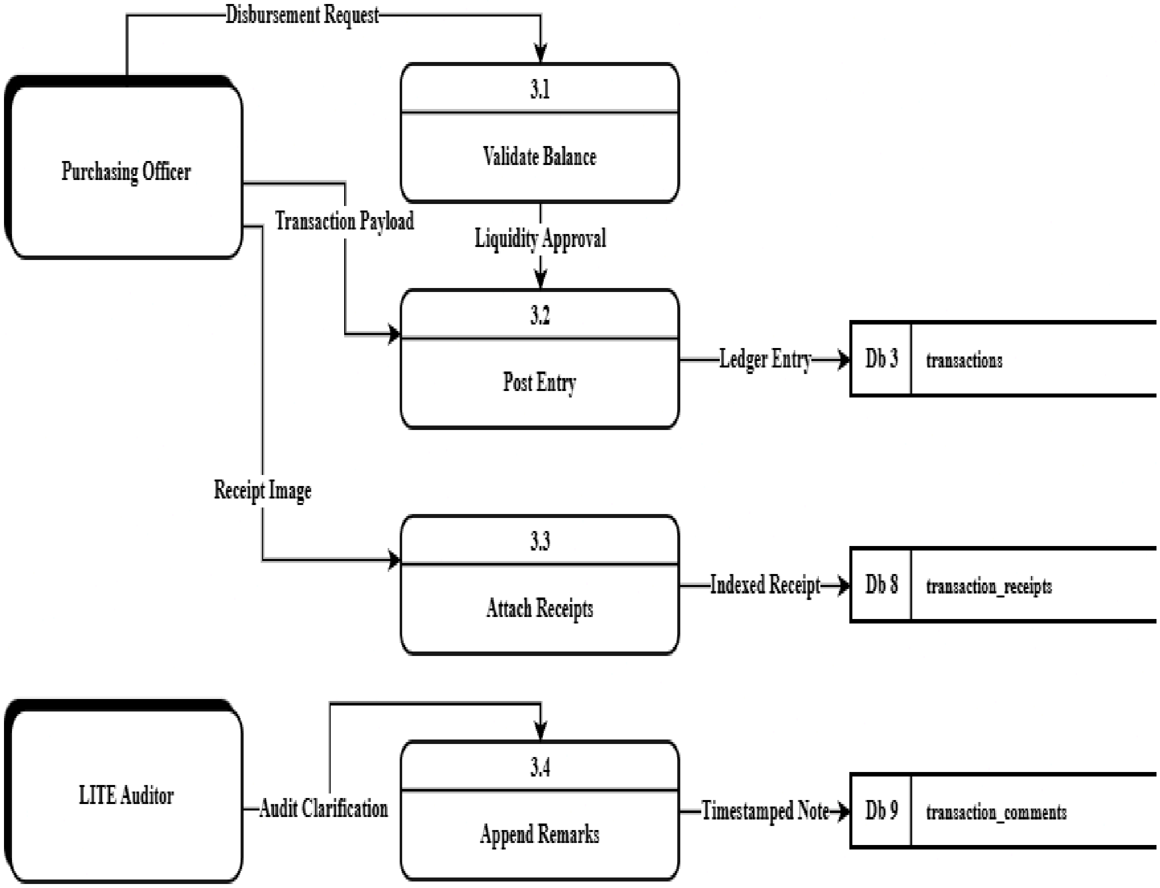

Disbursement Request
Purchasing Officer
3.1
Validate Balance
Transaction Payload
Liquidity Approval
3.2
Post Entry
Ledger Entry
Db 3
transactions
Receipt Image
3.3
Attach Receipts
Indexed Receipt
Db 8
transaction_receipts
LITE Auditor
Audit Clarification
3.4
Append Remarks
Timestamped Note
Db 9
transaction_comments

Figure 7: Diagram 1 of Process 3: Record Transactions

Figure 7 details the internal mechanisms of Process 3 (Record Transactions) across four specialized sub-processes governing financial entries, image proof indexing, and auditor collaboration.

When an operational expenditure is proposed, Sub-process 3.1 (Validate Balance) receives a Disbursement Request from the an Officer and evaluates requested funds against liquid currency in the cashbox; if funds are sufficient, it issues Liquidity Approval to Sub-process 3.2 (Post Entry). Concurrently, the purchasing officer submits the operational Transaction Payload (amount, date, category, narrative remarks), enabling Sub-process 3.2 to write a verified Ledger Entry to Db 3 (transactions).

To satisfy collegiate evidentiary standards, Sub-process 3.3 (Attach Receipts) captures uploaded Receipt Image proofs (commercial invoices or handwritten petty cash slips for tricycle fares), applies client-side compression and commits the Indexed Receipt into Db 8 (transaction_receipts). To facilitate transparent post-audit reviews without altering verified figures, Sub-process 3.4 (Append Remarks) allows the LITE Auditor to submit an Audit Clarification, persisting a Timestamped Note into Db 9 (transaction_comments).

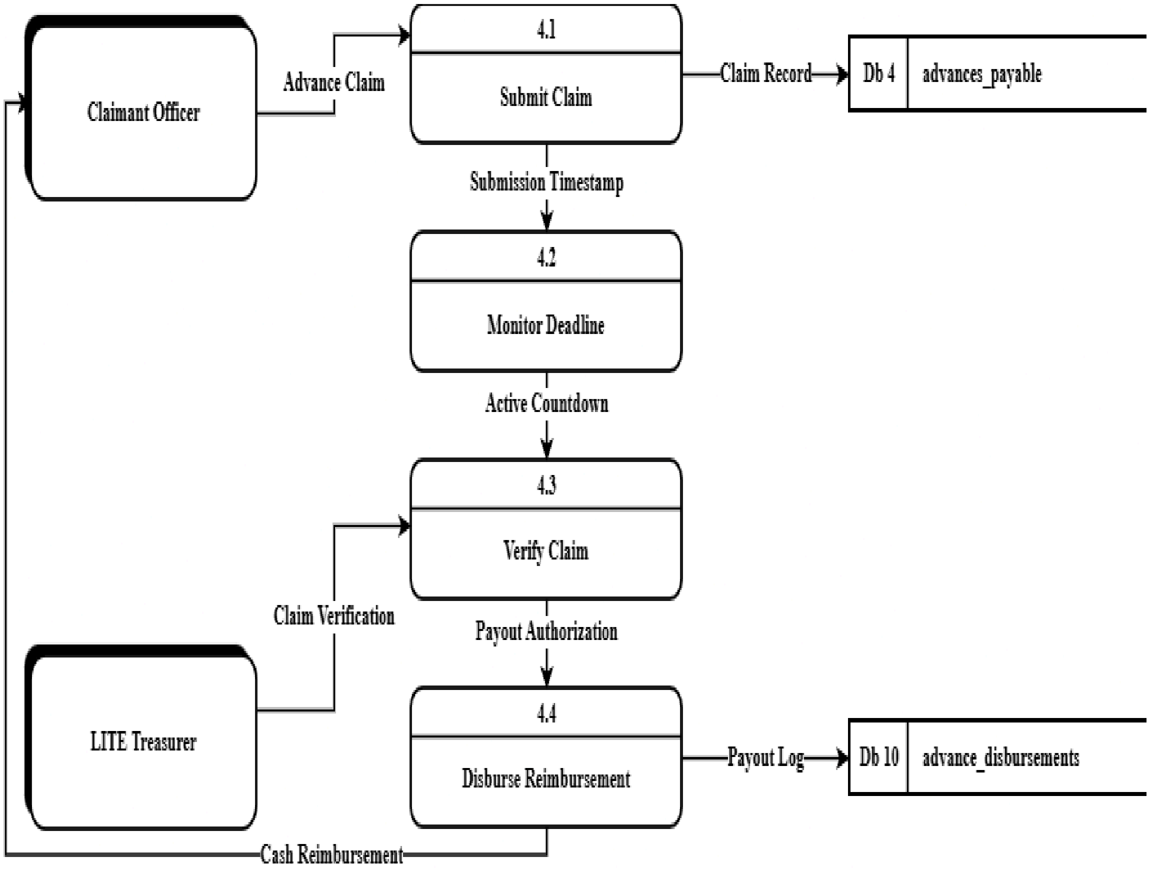

Claimant Officer
Advance Claim
4.1
Submit Claim
Claim Record
Db 4
advances_payable
Submission Timestamp
4.2
Monitor Deadline
Active Countdown
4.3
Verify Claim
Claim Verification
Payout Authorization
LITE Treasurer
4.4
Disburse Reimbursement
Payout Log
Db 10
advance_disbursements
Cash Reimbursement

Figure 8: Diagram 1 of Process 4: Manage Advances

Figure 8 depicts the workflow of Process 4 (Manage Advances), which regulates the handling of out-of-pocket organizational expenditures (abono) from initial submission through to cash reimbursement.

The procedure begins with Sub-process 4.1 (Submit Claim), where a faculty member or officer who expended personal funds logs an Advance Claim along with uploaded receipt images; this entry establishes an accounts payable liability record (Claim Record) within Db 4 (advances_payable) and forwards a Submission Timestamp to Sub-process 4.2 (Monitor Deadline). Next, under Sub-process 4.3 (Verify Claim), the LITE Treasurer examines the uploaded proofs to confirm receipt validity and issues a Claim Verification, which triggers a Payout Authorization for Sub-process 4.4 (Disburse Reimbursement). In instances where physical cash reserves are limited, Sub-process 4.4 accommodates partial installment payments, logging each transaction in Db 10 (advance_disbursements) as a Payout Log and disbursing physical Cash Reimbursement to the Claimant Officer (Executive Officers). This process progressively reduces the outstanding liability balance in Db 4 until the total advanced amount is cleared.

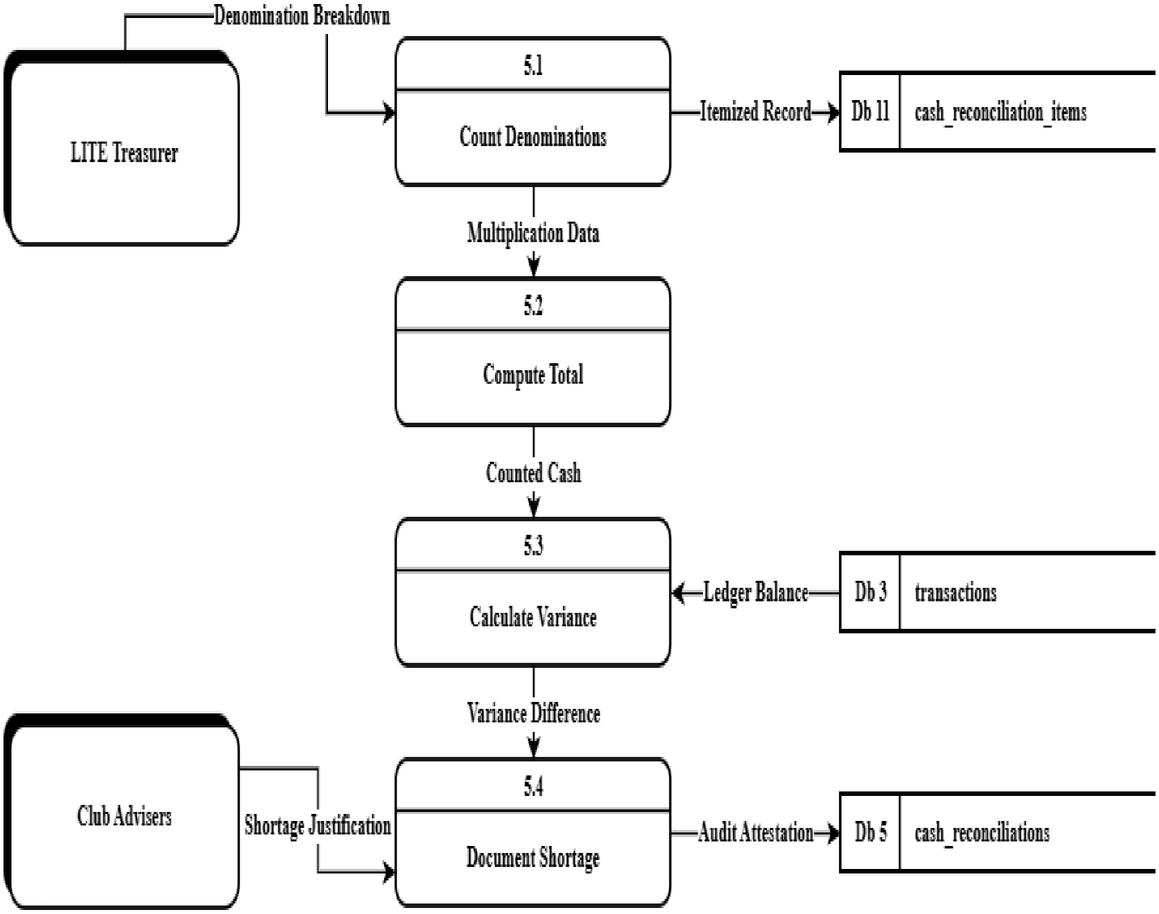

Denomination Breakdown
LITE Treasurer
5.1
Count Denominations
Itemized Record
Db 11
cash_reconciliation_items
Multiplication Data
5.2
Compute Total
Counted Cash
5.3
Calculate Variance
Ledger Balance
Db 3
transactions
Variance Difference
Club Advisers
Shortage Justification
5.4
Document Shortage
Audit Attestation
Db 5
cash_reconciliations

Figure 9: Diagram 1 of Process 5: Reconcile Cash

Figure 9 presents the architectural breakdown of Process 5 (Reconcile Cash), coordinating physical currency audits conducted at the adviser’s desk inside the Program Director's Room.

In Sub-process 5.1 (Count Denominations), the Executive Officers (LITE Treasurer) physically counts cashbox currency and inputs piece-counts across all 11 legal tender Philippine denominations (banknotes: ₱1,000, ₱500, ₱200, ₱100, ₱50, ₱20; coins: ₱20, ₱10, ₱5, ₱1, and ₱0.25 centavos) as Denomination Breakdown, persisting itemized counts into Db 11 (cash_reconciliation_items). Sub-process 5.1 forwards Multiplication Data to Sub-process 5.2 (Compute Total), which computes the exact physical currency subtotal with integer-centavo precision, supplying Counted Cash figures to Sub-process 5.3 (Calculate Variance).

Sub-process 5.3 pulls the theoretical balance (Ledger Balance) from Db 3 (transactions) and computes the variance (Counted Physical Cash minus Theoretical Balance). If an arithmetic discrepancy occurs, the Variance Difference is routed to Sub-process 5.4 (Document Shortage), where an official Shortage Justification is drafted and endorsed by the Club Advisers (Adviser & Admin), committing a certified Audit Attestation into Db 5 (cash_reconciliations) under dual-officer attestation.

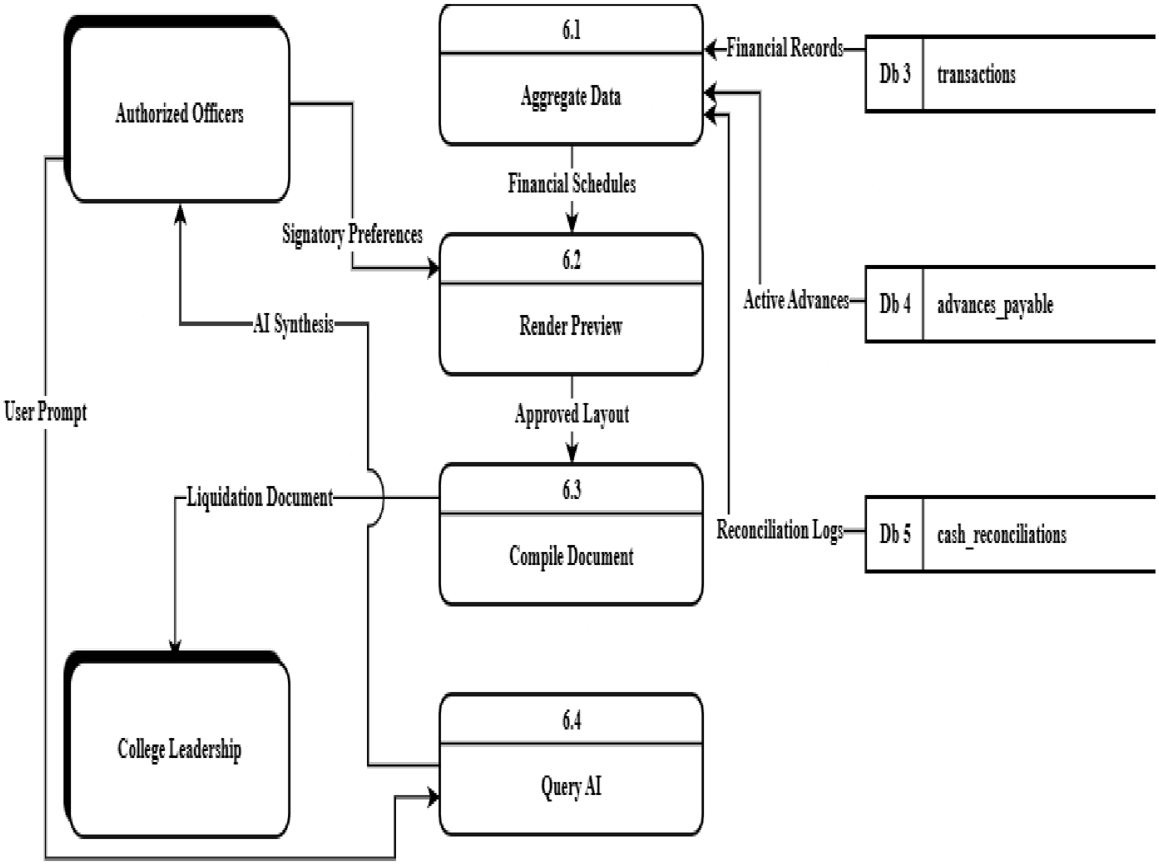

6.1
Aggregate Data
Financial Records
Db 3
transactions
Authorized Officers
Financial Schedules
Signatory Preferences
6.2
Render Preview
Active Advances
Db 4
advances_payable
AI Synthesis
User Prompt
Approved Layout
Liquidation Document
6.3
Compile Document
Reconciliation Logs
Db 5
cash_reconciliations
College Leadership
6.4
Query AI

Figure 10: Diagram 1 of Process 6: Compile Reports

Figure 10 outlines the Level 2 mechanisms of Process 6 (Compile Reports), managing automated liquidation synthesis and natural-language financial inquiry.

In Sub-process 6.1 (Aggregate Data), the reporting engine extracts verified financial datasets: Financial Records from Db 3 (transactions), Active Advances from Db 4 (advances_payable), and Reconciliation Logs from Db 5 (cash_reconciliations), compiling scoped Financial Schedules. Sub-process 6.2 (Render Preview) generates a pixel-accurate 1:1 live preview canvas incorporating adviser Signatory Preferences (toggling required administrative signatures across Treasurer, Auditor, Advisers, Program Director, and Dean).

Upon confirmation, Sub-process 6.3 (Compile Document) exports the verified figures into a standardized Microsoft Word (.docx) liquidation packet (Liquidation Document) featuring Calibri 11pt typography, borderless financial tables, formal transmittal headers to Co-Curricular Affairs Coordinator Ms. Ligaya H. Estrella, and automated 2-column, 4-up receipt attachment sheets, transmitting the packet to College Leadership for institutional clearance. Concurrently, Sub-Process 6.4 executes conversational intelligence queries. Authorized officers—including the President, Vice President, Treasurer, and Club Advisers—dispatch natural language prompts into the conversational AI Co-Pilot, prompting the system to fetch data figures, calculate net liquidity positions, and return grounded financial syntheses without risk of hallucination.

2.1.3 Use Case Diagram

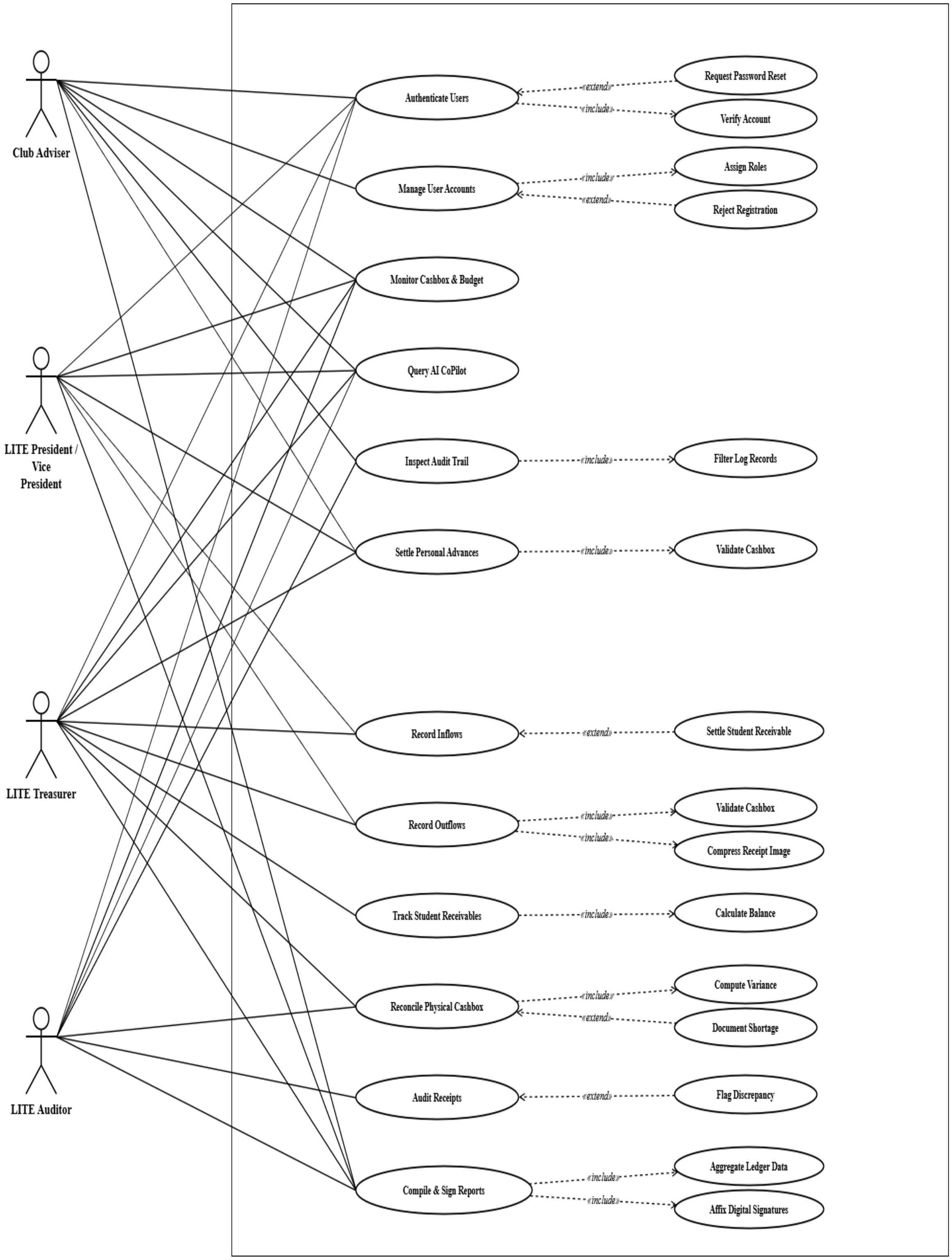

Club Adviser
LITE President / Vice President
LITE Treasurer
LITE Auditor
Authenticate Users
«extend»
Request Password Reset
«include»
Verify Account
Manage User Accounts
«include»
Assign Roles
«extend»
Reject Registration
Monitor Cashbox & Budget
Query AI CoPilot
Inspect Audit Trail
«include»
Filter Log Records
Settle Personal Advances
«include»
Validate Cashbox
Record Inflows
«extend»
Settle Student Receivable
Record Outflows
«include»
Validate Cashbox
«include»
Compress Receipt Image
Track Student Receivables
«include»
Calculate Balance
Reconcile Physical Cashbox
«include»
Compute Variance
«extend»
Document Shortage
Audit Receipts
«extend»
Flag Discrepancy
Compile & Sign Reports
«include»
Aggregate Ledger Data
«include»
Affix Digital Signatures

Figure 11: Use Case Diagram of FinLITE System

1. Authenticate Users

1.1 Introduction: Authorized officers and club advisers securely authenticate into the FinLITE web platform using institutional Google accounts or email credentials to access role-specific financial tools.

1.2 Actor: Club Adviser, LITE President / VP, LITE Treasurer, LITE Auditor

1.3 Pre-Condition: The user must possess an authorized Google account ending with @gmail.com and have access to the FinLITE login interface.

1.4 Post-Condition: An encrypted user session is initiated, assigned operational permissions are loaded, and the user is redirected to their designated dashboard.

1.5 Basic Flow:

1.5.1 The user navigates to the FinLITE web authentication gateway.

1.5.2 The user chooses to log in via Google OAuth 2.0 or enters registered email credentials and password.

1.5.3 The system validates the email domain format and verifies the credentials against the database.

1.5.4 The system checks that the account has been approved and activated by the Club Adviser (Verify Account).

1.5.5 The system generates an encrypted JSON Web Token (JWT) session containing the assigned user role.

1.5.6 The system directs the authenticated user to their role-specific executive or officer dashboard.

1.6 Alternative Flow:

1.6.1 If the user has forgotten their password, the user selects "Request Password Reset" to receive a secure recovery link via registered email.

1.6.2 If the user account is newly registered and pending approval, the system restricts access and displays a waiting notice directing them to contact the Club Adviser.

1.6.3 If the user remains inactive for 15 minutes, the system terminates the session and redirects the user to the login prompt.

2. Manage User Accounts

2.1 Introduction: The Club Adviser manages officer profiles, approves pending student accounts, assigns operational system roles, and revokes unauthorized access.

2.2 Actor: Club Adviser

2.3 Pre-Condition: The Club Adviser must be logged in with master administrative access rights.

2.4 Post-Condition: Officer profiles are updated, assigned roles and permissions are committed to the users database, and unauthorized applicants are removed.

2.5 Basic Flow:

2.5.1 The Club Adviser navigates to the User Management module from the administrative dashboard.

2.5.2 The system displays the roster of active officers and pending account registrations.

2.5.3 The Club Adviser selects a pending registrant and reviews their student credentials.

2.5.4 The Club Adviser selects "Assign Roles" and designates the approved operational role (President, Vice President, Treasurer, or Auditor).

2.5.5 The system enforces role mutual exclusion to ensure an individual cannot hold conflicting duties (e.g., simultaneous Treasurer and Auditor).

2.5.6 The Club Adviser confirms the activation, and the system updates the user record and unlocks platform access.

2.6 Alternative Flow:

2.6.1 If a registration is unrecognized or unverified, the Club Adviser selects "Reject Registration" to decline the application and purge the unverified record.

2.6.2 If an outgoing officer ends their academic term, the Club Adviser deactivates or archives the profile to revoke system privileges.

3. Monitor Cashbox & Budget

3.1 Introduction: Officers and advisers monitor live organizational financial liquidity, semester budget allocations, outstanding student receivables, and cashbox balances in real time.

3.2 Actor: Club Adviser, LITE President / VP, LITE Treasurer, LITE Auditor

3.3 Pre-Condition: The user must be authenticated with active system access.

3.4 Post-Condition: Real-time financial metric cards, visual spending graphs, and active liquidity summaries are compiled and presented on screen.

3.5 Basic Flow:

3.5.1 The user opens the FinLITE executive dashboard.

3.5.2 The system automatically retrieves verified entries from the single-entry ledger database, pulls physical cash counts from recent denomination audits, and gathers pending liabilities from the advance and receivable logs.

3.5.3 The system computes current physical cash on hand, net liquidity, total semester inflows, and total disbursements.

3.5.4 The dashboard renders visual balance cards, categorized expense progress bars, and recent activity logs.

3.5.5 The user reviews budget utilization metrics and monitors available cash reserves for planned student activities.

3.6 Alternative Flow:

3.6.1 If unrecorded or pending transactions cause unexpected variance warnings, the system displays an alert badge prompting custodians to conduct a cash reconciliation.

4. Query AI Co-Pilot

4.1 Introduction: Authorized leaders and advisers use the integrated conversational AI assistant to ask natural-language questions regarding organizational finances, budget status, and spending trends.

4.2 Actor: Club Adviser, LITE President / VP, LITE Treasurer, LITE Auditor

4.3 Pre-Condition: The user must be logged in with authorized access to the conversational assistant interface.

4.4 Post-Condition: Live ledger records are analyzed, and a grounded, read-only natural-language financial summary is returned without altering database data.

4.5 Basic Flow:

4.5.1 The user opens the AI Co-Pilot modal from the application header.

4.5.2 The user inputs a natural-language financial inquiry (e.g., "What is our remaining cash balance for event supplies?").

4.5.3 The system automatically fetches real-time balances, transaction logs, and receivable states directly from the single-entry ledger and relational database stores.

4.5.4 The system constructs a secure, structured context prompt and transmits it to the Google Gemini API.

4.5.5 The system displays the generated response with exact figures, dates, and category breakdowns.

4.6 Alternative Flow:

4.6.1 If the user enters a prompt unrelated to organizational finances or requests database modifications, the system returns a polite boundary notice explaining that the AI Co-Pilot is strictly a read-only financial query tool.

4.6.2 If network disruption interrupts the external API connection, the system alerts the user and falls back to standard dashboard financial summaries.

5. Inspect Audit Trail

5.1 Introduction: The Auditor and Club Adviser inspect immutable chronological logs of all database modifications, user logins, and financial entries to ensure transparency and accountability.

5.2 Actor: Club Adviser, LITE Auditor

5.3 Pre-Condition: The user must be logged in with audit inspection or administrative privileges.

5.4 Post-Condition: A tamper-evident chronological audit history is compiled, filtered, and presented for institutional verification.

5.5 Basic Flow:

5.5.1 The user accesses the Audit Trail module from the navigation sidebar.

5.5.2 The system queries the append-only audit log data store.

5.5.3 The user selects "Filter Log Records" to narrow entries by date range, action type (INSERT, UPDATE, DELETE), or officer ID.

5.5.4 The system renders the filtered log records detailing exact timestamps, acting user emails, target entities, and previous versus updated values.

5.5.5 The user evaluates the entries to verify regulatory compliance and confirm that no unauthorized modifications occurred.

5.6 Alternative Flow:

5.6.1 If no log events correspond to the applied filter parameters, the system displays an empty state notification stating that no matching audit records were found.

6. Settle Personal Advances

6.1 Introduction: Officers and advisers log, track, and liquidate out-of-pocket frontings (abono) incurred for official organizational purchases to obtain reimbursement from the physical cashbox.

6.2 Actor: Club Adviser, LITE President / VP, LITE Treasurer

6.3 Pre-Condition: An officer or adviser has expended personal funds for authorized club business and obtained proof of purchase.

6.4 Post-Condition: The advance liquidation claim is recorded and cash reimbursement is disbursed once verified.

6.5 Basic Flow:

6.5.1 The purchasing officer or adviser navigates to the Personal Advances module.

6.5.2 The actor logs a new advance claim by entering the purchase purpose, amount spent, date, and uploading the official receipt image.

6.5.3 The system registers the pending claim into the advances payable ledger.

6.5.4 The LITE President/VP reviews and endorses the liquidation claim.

6.5.5 The Treasurer selects "Validate Cashbox" to verify that sufficient physical cash is available to fund the reimbursement without overdraft.

6.5.6 The Treasurer disburses the physical cash to the claimant and marks the advance claim as settled.

6.6 Alternative Flow:

6.6.1 If the physical cashbox balance is insufficient to cover the full advance reimbursement, LITE allows partial cash payouts from available cashbox liquidity while logging the remaining balance as an active pending liability under Accounts Payable until further collections are received.

6.6.2 If an advance claim remains unliquidated as the 7-day countdown nears expiration, the system issues visual reminder alerts on the executive dashboard.

7. Record Inflows

7.1 Introduction: The Treasurer or designated officer records cash collections, event entry fees, tournament registration fees, batch merchandise sales, and booth rental fees (arkila) into the organizational single-entry ledger under Inflows.

7.2 Actor: LITE Treasurer, LITE President / VP

7.3 Pre-Condition: Physical cash has been collected and deposited into the physical cashbox.

7.4 Post-Condition: The cash inflow transaction is committed to the database, the cashbox balance increases, and a digital collection receipt is generated.

7.5 Basic Flow:

7.5.1 The designated officer navigates to the Transaction Recording module and selects "Record Inflow."

7.5.2 The actor enters the collection date, category (e.g., Event Entry Fee, Merchandise Pre-Order, Booth Rental Fee / Arkila, or Carried-Over Surplus), payer name, and cash amount received.

7.5.3 If the payment applies toward an outstanding student balance, the system invokes "Settle Student Receivable" to deduct the installment from the student's debt record.

7.5.4 The actor confirms the transaction details and submits the form.

7.5.5 The system updates the transactions ledger, increments the calculated cashbox balance, and generates an official collection receipt record.

7.6 Alternative Flow:

7.6.1 If an invalid, zero, or negative monetary amount is entered, the system displays an error prompt and prevents submission until a valid positive figure is provided.

8. Record Outflows

8.1 Introduction: The Treasurer or designated executive records operational expenses and cash disbursements paid directly from the physical cashbox, accompanied by digitized receipt proofs.

8.2 Actor: LITE Treasurer, LITE President / VP

8.3 Pre-Condition: The officer is authorized to disburse funds from the physical cashbox for verified organization expenses.

8.4 Post-Condition: The cash outflow is recorded, physical cashbox reserves are decremented, and an optimized receipt image is attached to the transaction record.

8.5 Basic Flow:

8.5.1 The designated officer opens the Transaction Recording module and selects "Record Outflow."

8.5.2 The actor inputs the payee name, expense category, disbursement date, purpose, and amount paid.

8.5.3 The system executes "Validate Cashbox" to ensure that the required disbursement amount does not exceed the current physical cashbox balance.

8.5.4 The actor uploads the photographic receipt proof, and the system executes "Compress Receipt Image" to downsample the file below 1 MB.

8.5.5 The actor confirms the disbursement, and the system deducts the amount from the cashbox balance and logs the transaction.

8.6 Alternative Flow:

8.6.1 If the disbursement amount exceeds available cashbox funds, the system triggers a cash deficit alert and blocks transaction submission to prevent cashbox overdraft.

8.6.2 If the uploaded receipt image format is unsupported or unreadable, the system prompts the user to upload a valid JPEG, PNG, or WebP document image.

9. Track Student Receivables

9.1 Introduction: The Treasurer monitors student merchandise installment plans (hulog), tracks unpaid dues, and updates personal debt ledgers upon receiving partial payments.

9.2 Actor: LITE Treasurer

9.3 Pre-Condition: A student member has acquired organization merchandise or apparel with an authorized unpaid or installment balance.

9.4 Post-Condition: The installment payment is recorded, the remaining student debt is updated, and fully paid accounts are marked as cleared.

9.5 Basic Flow:

9.5.1 The Treasurer navigates to the Student Receivables module.

9.5.2 The Treasurer searches for the debtor student by student identification number or surname.

9.5.3 The system displays the student's active debt ledger, merchandise items acquired, and payment history.

9.5.4 The Treasurer logs the received installment amount and payment date.

9.5.5 The system executes "Calculate Balance" to subtract the payment from the remaining balance and displays the updated balance.

9.5.6 The system logs the payment into the accounts receivable ledger and marks the account as fully settled when the balance reaches zero.

9.6 Alternative Flow:

9.6.1 If the entered installment amount exceeds the total remaining debt, the system alerts the Treasurer and prevents overpayment entry.

10. Reconcile Physical Cashbox

10.1 Introduction: The Treasurer and Auditor jointly count physical cash inside the metal cashbox and reconcile physical currency counts against the digital system ledger.

10.2 Actor: LITE Treasurer, LITE Auditor

10.3 Pre-Condition: The Treasurer and Auditor are physically present at the metal cashbox within the Program Director's office.

10.4 Post-Condition: The denomination breakdown is saved, the physical cash total is computed, and variance against ledger balances is recorded.

10.5 Basic Flow:

10.5.1 The Treasurer and Auditor open the Cashbox Reconciliation module.

10.5.2 The officers unlock the metal cashbox and count physical bills and coins.

10.5.3 The Treasurer enters physical piece counts for all 11 Philippine bill and coin denominations on the digital keypad.

10.5.4 The system computes the total physical cash on hand and executes "Compute Variance" against the recorded book balance.

10.5.5 The Auditor reviews the denomination counts, verifies the physical cash total, and co-signs the reconciliation session.

10.5.6 The system saves the reconciliation log and locks the record against post-audit alteration.

10.6 Alternative Flow:

10.6.1 If a discrepancy exists between counted physical cash and the recorded ledger balance, the system invokes "Document Shortage," requiring custodians to input a written justification before finalizing the session.

11. Audit Receipts

11.1 Introduction: The Auditor conducts independent, read-only reviews of uploaded receipt images and expense proofs to verify compliance with institutional spending guidelines.

11.2 Actor: LITE Auditor

11.3 Pre-Condition: Financial disbursement transactions or advance liquidations with attached receipts have been recorded in the system.

11.4 Post-Condition: Receipts are marked as verified, or non-compliant receipts are tagged with formal audit observation notices.

11.5 Basic Flow:

11.5.1 The Auditor opens the Receipt Audit module from the navigation sidebar.

11.5.2 The system displays a queue of recorded disbursements and expense transactions.

11.5.3 The Auditor opens an item and inspects the receipt image in the high-resolution viewer.

11.5.4 The Auditor verifies vendor authenticity, receipt date, itemized items, and total amount against the recorded transaction figures.

11.5.5 The Auditor confirms compliance and stamps the receipt as verified.

11.6 Alternative Flow:

11.6.1 If a receipt is illegible, missing required details, or mismatches the recorded amount, the Auditor invokes "Flag Discrepancy" to attach an audit finding notice and alert custodians to provide clarification.

12. Compile & Sign Reports

12.1 Introduction: Authorized officers and advisers compile comprehensive financial liquidation packets, review schedules, sequentially affix digital signatures, and export official dossiers.

12.2 Actor: Club Adviser, LITE President / VP, LITE Treasurer, LITE Auditor

12.3 Pre-Condition: All financial transactions for the reporting cycle are recorded, receipts are audited, and physical cash is reconciled.

12.4 Post-Condition: A standardized liquidation packet containing financial schedules, digitized receipts, and verified signatures is exported as a Microsoft Word (.docx).

12.5 Basic Flow:

12.5.1 The Treasurer accesses the Financial Reports module and selects the active reporting period.

12.5.2 The system executes "Aggregate Ledger Data," pulling verified records from transactions, cash reconciliations, advances, and student dues.

12.5.3 The system renders a 1:1 pixel-accurate live preview of the liquidation report.

12.5.4 The system executes "Affix Digital Signatures," routing the report through sequential review by the Treasurer, Auditor, President, and Club Advisers.

12.5.5 Upon collecting all required digital sign-offs, the Treasurer initiates report export.

12.5.6 The system generates a compliant Microsoft Word (.docx) dossier with Calibri 11pt typography, borderless tables, transmittal headers, and 4-up receipt attachment sheets.

12.6 Alternative Flow:

12.6.1 If an authorized signatory identifies an unverified transaction or missing receipt during review, they reject the packet with explanatory comments, returning it to draft status for correction.

2.1.4 Activity Diagram

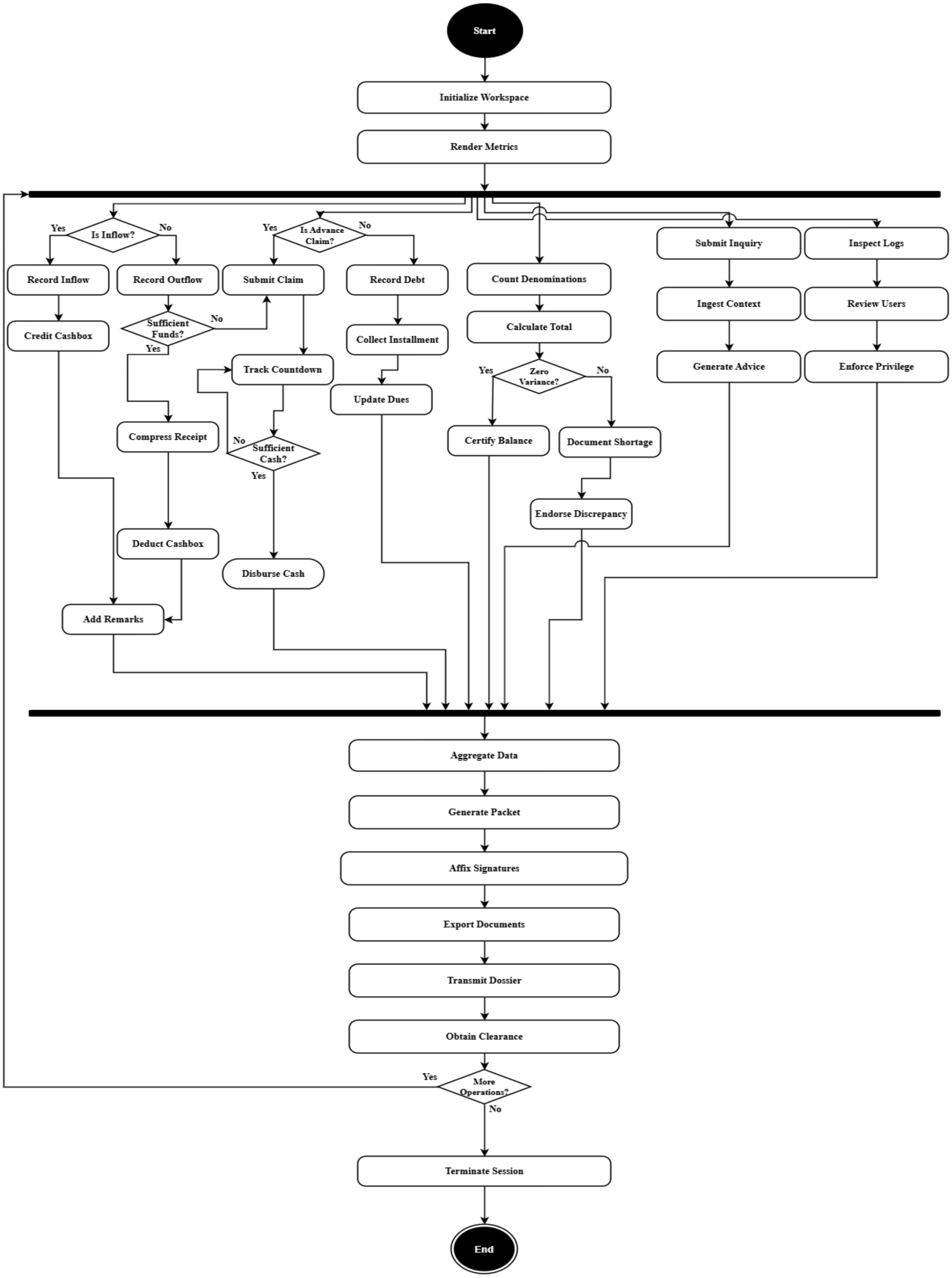

Start
Initialize Workspace
Render Metrics
Yes
Is Inflow?
No
Yes
Is Advance Claim?
No
Submit Inquiry
Inspect Logs
Record Inflow
Record Outflow
Submit Claim
Record Debt
Count Denominations
Ingest Context
Review Users
Credit Cashbox
Sufficient Funds?
No
Yes
Collect Installment
Calculate Total
Generate Advice
Enforce Privilege
Track Countdown
Yes
Zero Variance?
No
Update Dues
Compress Receipt
No
Sufficient Cash?
Yes
Certify Balance
Document Shortage
Endorse Discrepancy
Deduct Cashbox
Disburse Cash
Add Remarks
Aggregate Data
Generate Packet
Affix Signatures
Export Documents
Transmit Dossier
Obtain Clearance
Yes
More Operations?
No
Terminate Session
End

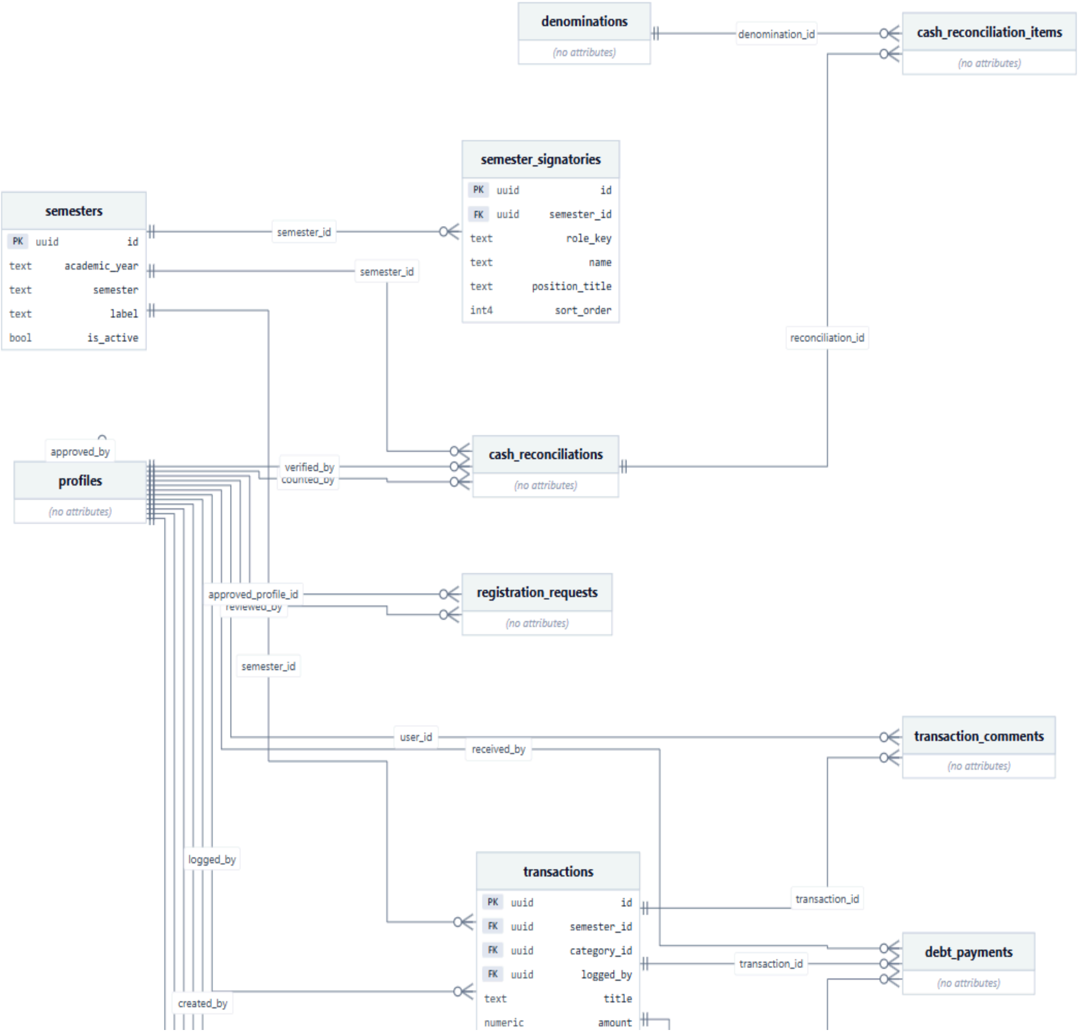
Figure 12: Activity Diagram of FinLITE System

The Activity Diagram illustrates the operational architecture of FinLITE. Starting from system entry, the platform initializes the workspace and renders live metrics before splitting into five parallel tracks: transaction management, advances and receivables, cashbox reconciliation, AI assistance, and audit governance. Each workflow enforces strict policy rules, such as overdraft checks and non-volatile logging. Finally, the parallel processes merge to compile validated records into compliant liquidation packages, gather required electronic signatures, and route documentation for institutional review.

2.2 Project Design

2.2.1 Entity Relationship Diagram

The entity relationship diagram shows the connection of primary keys and foreign keys of tables that are used in FinLITE.


denominations
(no attributes)
denomination_id
cash_reconciliation_items
(no attributes)
semester_signatories
PK uuid id
FK uuid semester_id
text role_key
text name
text position_title
int4 sort_order
semesters
PK uuid id
text academic_year
text semester
text label
bool is_active
semester_id
semester_id
reconciliation_id
approved_by
profiles
(no attributes)
verified_by
counted_by
cash_reconciliations
(no attributes)
approved_profile_id
reviewed_by
registration_requests
(no attributes)
semester_id
user_id
received_by
transaction_comments
(no attributes)
logged_by
transactions
PK uuid id
FK uuid semester_id
FK uuid category_id
FK uuid logged_by
text title
numeric amount
transaction_id
transaction_id
debt_payments
(no attributes)
created_by

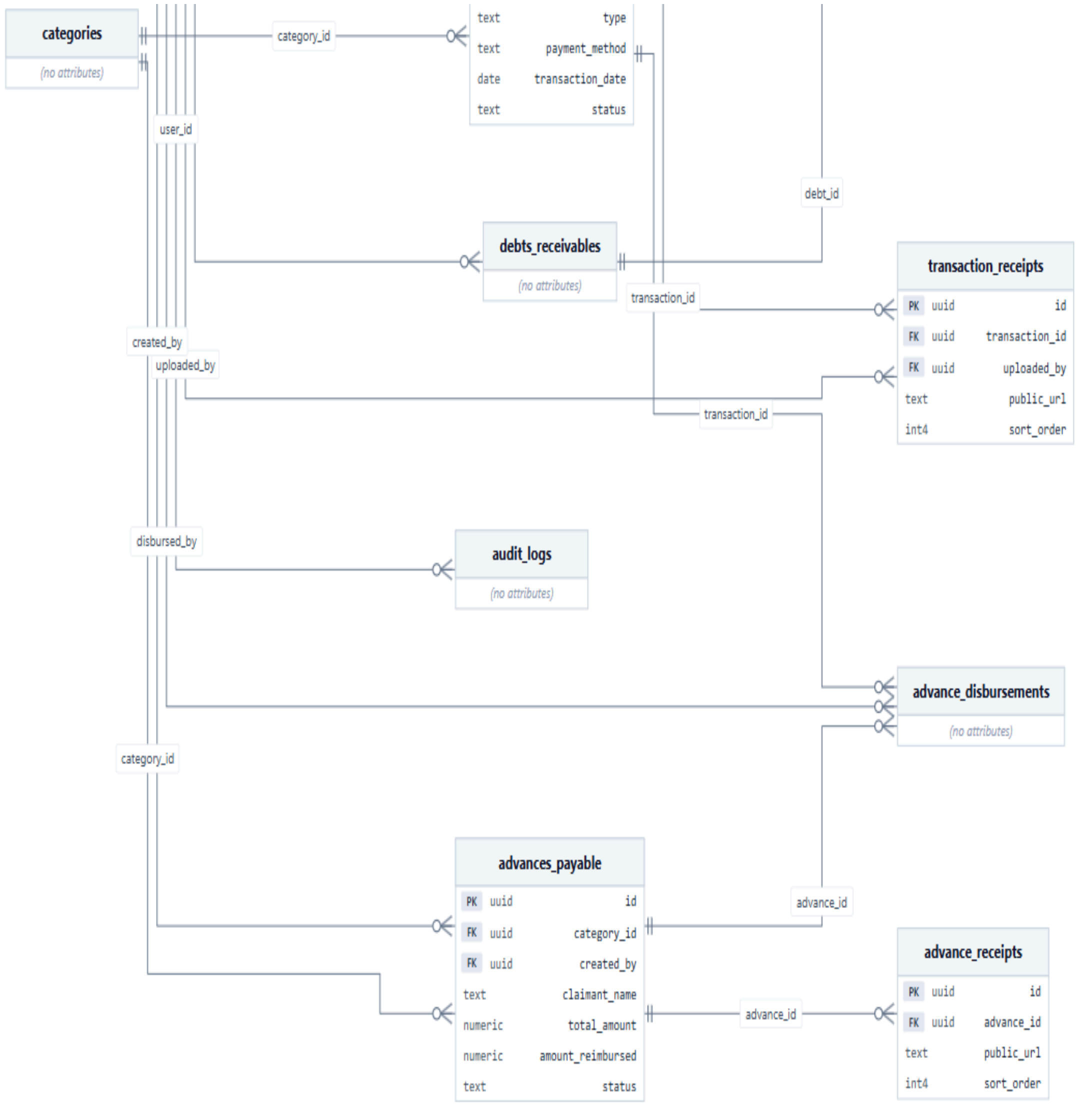

categories
(no attributes)
category_id
text type
text payment_method
date transaction_date
text status
user_id
debt_id
debts_receivables
(no attributes)
transaction_receipts
transaction_id
PK uuid id
FK uuid transaction_id
FK uuid uploaded_by
text public_url
int4 sort_order
created_by
uploaded_by
transaction_id
disbursed_by
audit_logs
(no attributes)
advance_disbursements
(no attributes)
category_id
advances_payable
PK uuid id
FK uuid category_id
FK uuid created_by
text claimant_name
numeric total_amount
numeric amount_reimbursed
text status
advance_id
advance_receipts
PK uuid id
FK uuid advance_id
text public_url
int4 sort_order
advance_id

Figure 13: Entity Relationship Diagram

Figure 13 illustrates the Third Normal Form (3NF) Entity-Relationship Diagram of FinLITE, modeling the structural relationships, cardinality, and constraints across the system’s 17 core relational tables. The schema is organized into five functional subsystems: the central financial ledger with digitized multi-receipt attachments, physical cashbox denomination audits, accounts payable for officer advances (abono), student receivables, and academic period governance with official institutional signatory chains. Referential integrity is strictly maintained through UUID primary and foreign key constraints, eliminating data redundancy, preventing anomaly risks, and ensuring point-in-time arithmetical accuracy for formal College of Computer Studies (CCS) financial liquidation reports.

2.2.2 Data Dictionary

The data dictionary contains the collection and description of data in the database. It comprises the field name, field type, required, key, and domain

Table 1: Profiles

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the profile record.
auth_user_id	uuid	YES	UK	The foreign ID referencing auth.users.
full_name	varchar(150)	YES		The full name of the user.
email	varchar(255)	YES	UK	The email address of the user.
role	varchar(20)	YES		The system role of the user.
status	varchar(20)	YES		The account approval status.
can_record_inflows	boolean	YES		Flag allowing inflow entry.

can_record_outflows	boolean	YES		Flag allowing outflow entry.
can_edit_transactions	boolean	YES		Flag allowing transaction edits.
can_audit_transactions	boolean	YES		Flag allowing transaction audits.
can_manage_debts	boolean	YES		Flag allowing debt management.
can_manage_advances	boolean	YES		Flag allowing advance management.
can_reconcile_cash	boolean	YES		Flag allowing cash reconciliation.
can_generate_reports	boolean	YES		Flag allowing report generation.
avatar_url	text	NO		The profile image URL.
created_at	timestamptz	YES		The date and time registered.

updated_at	timestamptz	YES		The date and time last modified.

Table 2: Registration_Requests

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the request.
first_name	varchar(100)	YES		The first name of the applicant.
last_name	varchar(100)	YES		The last name of the applicant.
email	varchar(255)	YES	UK	The email address of the applicant.
contact_number	varchar(20)	NO		The contact number of the applicant.
auth_provider	varchar(20)	YES		The sign-in method ('email' or 'google').
google_id	varchar(100)	NO		The Google subject identifier.

avatar_url	text	NO		The avatar photo URL.
status	varchar(20)	YES		The approval status ('pending', 'approved', 'rejected').
rejection_reason	text	NO		The reason for application rejection.
requested_role	varchar(20)	YES		The requested user role.
reviewed_by	uuid	NO	FK	The ID of the reviewing adviser.
reviewed_at	timestamptz	NO		The date and time reviewed.
created_at	timestamptz	YES		The date and time submitted.
updated_at	timestamptz	YES		The date and time last modified.

Table 3: Auth_Security_State

Field	Type/Length	Required	Key	Domain

auth_user_id	uuid	YES	PK	The user ID referencing auth.users.
password_enroll_until	timestamptz	NO		Expiration timestamp for setting password.
google_link_until	timestamptz	NO		Expiration timestamp for linking Google.
password_set_at	timestamptz	NO		The date and time password was set.
google_linked_at	timestamptz	NO		The date and time Google was linked.
updated_at	timestamptz	YES		The date and time last updated.

Table 4: Email_Verification_OTPs

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the OTP record.

email	varchar(255)	YES		The email address receiving the OTP.
code	varchar(6)	YES		The 6-digit verification code.
purpose	varchar(30)	YES		The purpose ('registration' or 'password_reset').
expires_at	timestamptz	YES		The expiration date and time.
used	boolean	YES		Flag indicating if code was used.
created_at	timestamptz	YES		The date and time code was created.

Table 5: Semesters

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the semester record.
academic_year	varchar(20)	YES		The academic school year.

semester	varchar(20)	YES		The semester term name.
label	varchar(100)	YES		The display label of the semester.
starting_cash_on_hand	decimal(12,2)	YES		The opening cashbox balance.
is_active	boolean	YES		Flag indicating the active semester.
created_at	timestamptz	YES		The date and time created.

Table 6: Semester_Signatories

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the signatory record.
academic_year	varchar(20)	YES		The academic school year.
semester	varchar(20)	YES		The semester term name.

role_key	varchar(50)	YES		The role identifier of the signatory.
name	varchar(150)	YES		The full name of the signatory.
position_title	varchar(150)	YES		The official institutional title.
category	varchar(20)	YES		The signatory tier ('OFFICER', 'ADVISER', 'ADMIN').
sort_order	int	YES		The display order on reports.
is_active	boolean	YES		Flag indicating if signatory is active.
created_at	timestamptz	YES		The date and time created.
updated_at	timestamptz	YES		The date and time last updated.

Table 7: Categories

Field	Type/Length	Required	Key	Domain

id	uuid	YES	PK	The unique ID of the category record.
name	varchar(100)	YES		The name of the category.
type	varchar(10)	YES		The category type ('INFLOW' or 'OUTFLOW').
description	text	NO		The description of the category.
created_at	timestamptz	YES		The date and time created.

Table 8: Transactions

Field	**Type/Length**	**Required**	**Key**	**Domain**
id	uuid	YES	PK	The unique ID of the transaction record.
semester_id	uuid	YES	FK	The semester ID referencing Semesters.
title	varchar(255)	YES		The title of the transaction.

description	text	NO		The description of the transaction.
amount	decimal(12,2)	YES		The monetary amount in pesos.
type	varchar(10)	YES		The transaction type ('INFLOW' or 'OUTFLOW').
payment_method	varchar(20)	YES		The payment method ('CASH').
category_id	uuid	NO	FK	The category ID referencing Categories.
logged_by	uuid	NO	FK	The user ID who encoded the entry.
transaction_date	date	YES		The date of the transaction.
status	varchar(20)	YES		The audit status ('PENDING', 'VERIFIED', 'REVISION_REQUIRED').
audited_by	uuid	NO	FK	The auditor ID who verified the entry.

audited_at	timestamptz	NO		The date and time audited.
is_reimbursement	boolean	YES		Flag indicating advance reimbursement.
payer_name	varchar(150)	NO		The name of the payer.
claimant_name	varchar(150)	NO		The name of the reimbursement claimant.
notes	text	NO		Additional transaction notes.
created_at	timestamptz	YES		The date and time created.
updated_at	timestamptz	YES		The date and time last updated.

Table 9: Transaction_Receipts

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the receipt record.

transaction_id	uuid	YES	FK	The transaction ID referencing Transactions.
public_url	text	YES		The public image URL of the receipt.
file_path	text	NO		The file path in cloud storage.
file_name	varchar(255)	NO		The original file name.
file_size	int	NO		The file size in bytes.
mime_type	varchar(50)	NO		The MIME type of the file.
sort_order	int	YES		The display sequence number.
created_at	timestamptz	YES		The date and time uploaded.

Table 10: Transaction_Comments

Field	Type/Length	Required	Key	Domain

id	uuid	YES	PK	The unique ID of the comment record.
transaction_id	uuid	YES	FK	The transaction ID referencing Transactions.
author_id	uuid	NO	FK	The user ID of the comment author.
comment_text	text	YES		The text content of the audit remark.
created_at	timestamptz	YES		The date and time commented.
updated_at	timestamptz	YES		The date and time last updated.

Table 11: Advances_Payable

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the advance claim.
semester_id	uuid	YES	FK	The semester ID referencing Semesters.

claimant_name	varchar(150)	YES		The name of the advance claimant.
claimant_contact	varchar(50)	NO		The contact details of the claimant.
title	varchar(255)	YES		The title of the advance claim.
category_id	uuid	NO	FK	The category ID referencing Categories.
event_name	varchar(150)	NO		The name of the associated event.
total_amount	decimal(12,2)	YES		The total advanced amount in pesos.
amount_reimbursed	decimal(12,2)	YES		The total reimbursed amount.
status	varchar(25)	YES		The settlement status ('UNPAID', 'PARTIALLY_REIMBURSED', 'SETTLED', 'CANCELLED').
advance_date	date	YES		The date of the advance.

notes	text	NO		Additional notes for the claim.
created_by	uuid	NO	FK	The user ID who filed the claim.
created_at	timestamptz	YES		The date and time created.
updated_at	timestamptz	YES		The date and time last updated.

Table 12: Advance_Disbursements

Field	**Type/Length**	**Required**	**Key**	**Domain**
id	uuid	YES	PK	The unique ID of the disbursement record.
advance_id	uuid	YES	FK	The advance ID referencing Advances_Payable.
transaction_id	uuid	NO	FK	The transaction ID referencing Transactions.
amount	decimal(12,2)	YES		The disbursement amount in pesos.

payment_method	varchar(20)	YES		The payment method ('CASH').
disbursement_date	timestamptz	YES		The date and time disbursed.
disbursed_by	uuid	NO	FK	The user ID who released the cash.
notes	text	NO		Additional disbursement notes.
created_at	timestamptz	YES		The date and time created.

Table 13: Advance_Receipts

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the receipt record.
advance_id	uuid	YES	FK	The advance ID referencing Advances_Payable.
public_url	text	YES		The public image URL of the receipt.

sort_order	int	YES		The display sequence number.
created_at	timestamptz	YES		The date and time uploaded.

Table 14: Debts_Receivables

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the receivable record.
semester_id	uuid	YES	FK	The semester ID referencing Semesters.
debtor_name	varchar(150)	YES		The full name of the debtor.
debtor_contact	varchar(50)	NO		The contact details of the debtor.
title	varchar(255)	YES		The description of items ordered.
total_amount	decimal(12,2)	YES		The total debt amount in pesos.
amount_paid	decimal(12,2)	YES		The total amount paid to date.

status	varchar(20)	YES		The settlement status ('UNPAID', 'PARTIALLY_PAID', 'SETTLED', 'CANCELLED').
due_date	date	NO		The due date for payment.
notes	text	NO		Additional receivable notes.
created_by	uuid	NO	FK	The user ID who logged the order.
created_at	timestamptz	YES		The date and time created.
updated_at	timestamptz	YES		The date and time last updated.

Table 15: Debt_Payments

Field	**Type/Length**	**Required**	**Key**	**Domain**
id	uuid	YES	PK	The unique ID of the payment record.
debt_id	uuid	YES	FK	The debt ID referencing Debts_Receivables.

transaction_id	uuid	NO	FK	The transaction ID referencing Transactions.
amount	decimal(12,2)	YES		The payment amount in pesos.
payment_method	varchar(20)	YES		The payment method ('CASH').
payment_date	timestamptz	YES		The date and time paid.
received_by	uuid	NO	FK	The user ID who received the cash.
notes	text	NO		Additional payment notes.
created_at	timestamptz	YES		The date and time created.

Table 16: Denominations

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the denomination.

code	varchar(30)	YES	UK	The unique denomination code.
label	varchar(50)	YES		The display label of the currency.
currency_type	varchar(10)	YES		The currency type ('BILL' or 'COIN').
face_value	decimal(12,2)	YES		The face value in pesos.
display_order	int	YES		The display order on keypad.
is_active	boolean	YES		Flag indicating if actively counted.
created_at	timestamptz	YES		The date and time created.

Table 17: Cash_Reconciliations

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the reconciliation session.

semester_id	uuid	YES	FK	The semester ID referencing Semesters.
reconciliation_date	timestamptz	YES		The date and time of reconciliation.
counted_by	uuid	NO	FK	The user ID who counted cash.
verified_by	uuid	NO	FK	The user ID who verified count.
physical_total	decimal(12,2)	YES		The total physical cash counted.
ledger_cash_balance	decimal(12,2)	YES		The theoretical ledger balance.
variance_amount	decimal(12,2)	YES		The variance difference amount.
variance_status	varchar(20)	YES		The status ('BALANCED', 'SHORTAGE', 'OVERAGE').
notes	text	NO		General reconciliation remarks.
adviser_approval_notes	text	NO		Adviser shortage justification notes.

status	varchar(20)	YES		The session status ('DRAFT', 'SUBMITTED', 'VERIFIED').
created_at	timestamptz	YES		The date and time created.

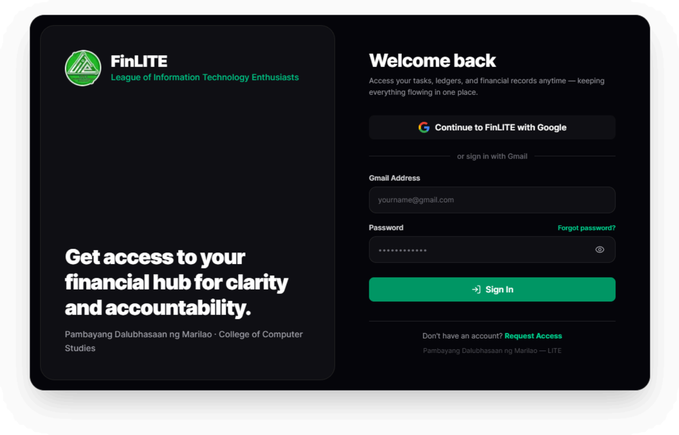
Table 18: Cash_Reconciliation_Items

Field	Type/Length	Required	Key	Domain
id	uuid	YES	PK	The unique ID of the item count.
reconciliation_id	uuid	YES	FK	The session ID referencing Cash_Reconciliations.
denomination_id	uuid	YES	FK	The denomination ID referencing Denominations.
quantity	int	YES		The piece-count quantity.
subtotal	decimal(12,2)	YES		The calculated currency subtotal.
created_at	timestamptz	YES		The date and time recorded.

2.3. Project Structure

2.3.1 Screen Layout/UI

The project design shows the different forms that can be seen in the graphical user interface (GUI) of the system.



FinLITE
League of Information Technology Enthusiasts

Get access to your financial hub for clarity and accountability.

Pambayang Dalubhasaan ng Marilao · College of Computer Studies

Welcome back

Access your tasks, ledgers, and financial records anytime — keeping everything flowing in one place.

Gmail Address: yourname@gmail.com

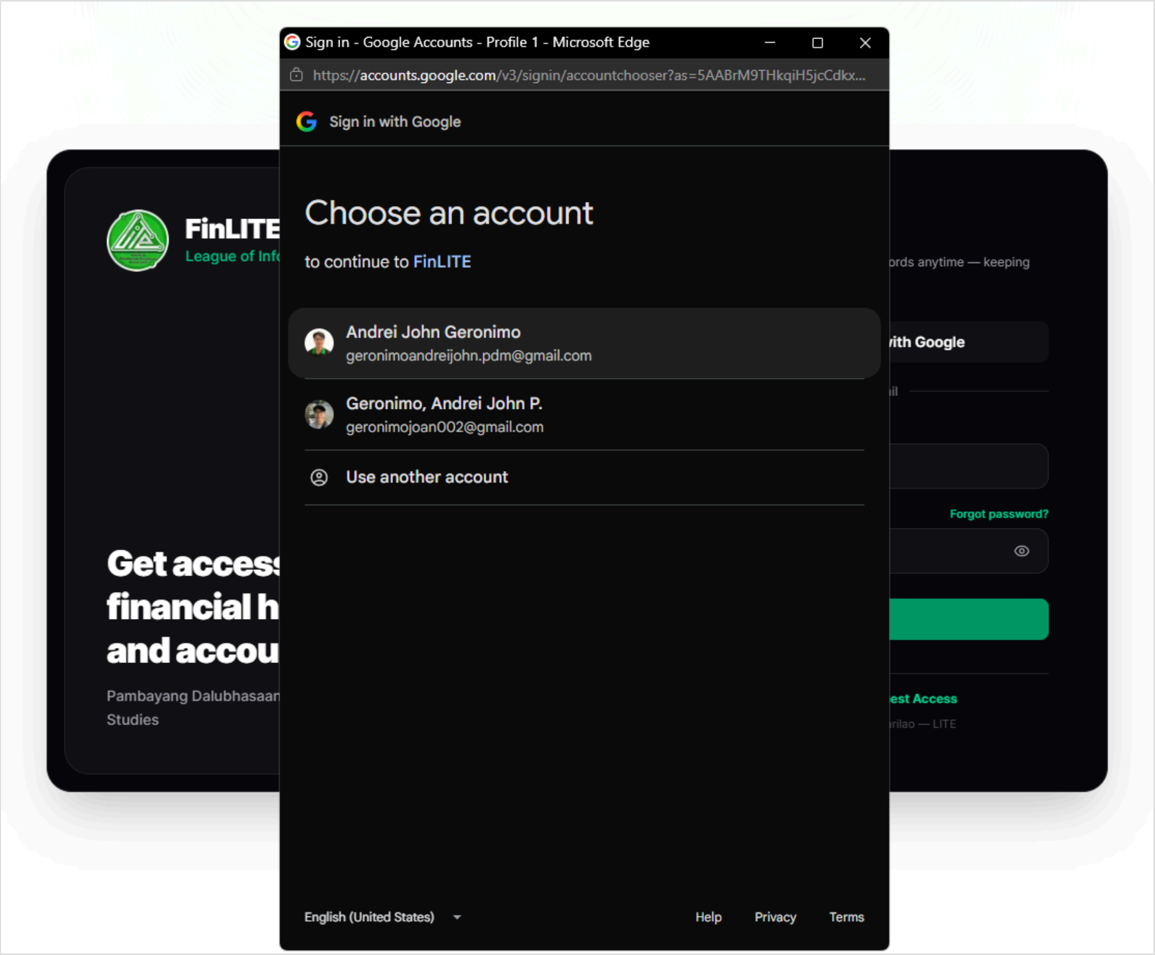
Password: ••••••••••••

Don't have an account? Request Access

Pambayang Dalubhasaan ng Marilao — LITE

Figure 14. User Sign In Page

This figure shows the authentication and sign-in page of the FinLITE platform, allowing registered student officers and faculty club advisers to access their operational workspace. It features a branded left overview panel emphasizing financial clarity and institutional accountability for Pambayang Dalubhasaan ng Marilao College of Computer Studies, alongside a right-side credential card containing a One-Tap Google Workspace OAuth sign-in button, input fields for institutional Gmail and password, a "Forgot password?" retrieval link, a "Sign In" button, and a "Request Access" registration link. Also here, the user can authenticate into the platform using their Google credentials or email and password, initiate secure password recovery, or navigate to the registration request form.



Sign in - Google Accounts - Profile 1 - Microsoft Edge

https://accounts.google.com/v3/signin/accountchooser?as=5AABrM9THkqiH5jcCdkx...

Sign in with Google

Choose an account

to continue to FinLITE

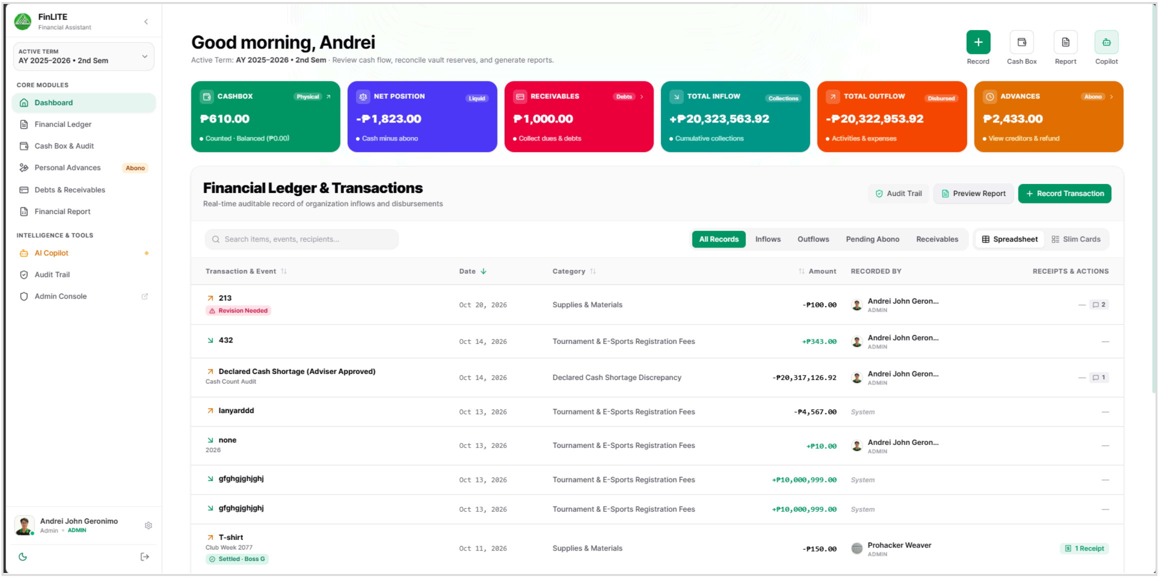
Andrei John Geronimo
geronimoandreijohn.pdm@gmail.com

Geronimo, Andrei John P.
geronimojoan002@gmail.com

Use another account
English (United States)
Help
Privacy
Terms
FinLITE
League of Info
Get access
financial h
and accou
Pambayang Dalubhasaan
Studies
ords anytime — keeping
ith Google
Forgot password?
est Access
rilao — LITE

Figure 15. Google OAuth 2.0 Single Sign-On Modal

This figure shows the institutional Google Identity Services authentication modal overlaying the FinLITE login screen. It displays active Google accounts associated with the device, complete with user profile avatars, full names, and registered Gmail addresses, providing a one-click Single Sign-On (SSO) mechanism. Also here, the officer can select their authorized Google account to achieve sub-millisecond, cryptographic authentication into the system without entering manual password strings.


FinLITE
Financial Assistant
ACTIVE TERM
AY 2025–2026 • 2nd Sem
CORE MODULES
Dashboard
Financial Ledger
Cash Box & Audit
Personal Advances
Abono
Debts & Receivables
Financial Report
INTELLIGENCE & TOOLS
AI Copilot
Audit Trail
Admin Console
Andrei John Geronimo
Admin • ADMIN
Record
Cash Box
Report
Copilot
Audit Trail
Preview Report
Record Transaction
Search items, events, recipients...
All Records
Inflows
Outflows
Pending Abono
Receivables
Spreadsheet
Slim Cards

Good morning, Andrei

Active Term: **AY 2025–2026 • 2nd Sem** · Review cash flow, reconcile vault reserves, and generate reports.

CASHBOX (Physical): ₱610.00 • Counted · Balanced (₱0.00)

NET POSITION (Liquid): -₱1,823.00 • Cash minus abono

RECEIVABLES (Debts): ₱1,000.00 • Collect dues & debts

TOTAL INFLOW (Collections): +₱20,323,563.92 • Cumulative collections

TOTAL OUTFLOW (Disbursed): -₱20,322,953.92 • Activities & expenses

ADVANCES (Abono): ₱2,433.00 • View creditors & refund

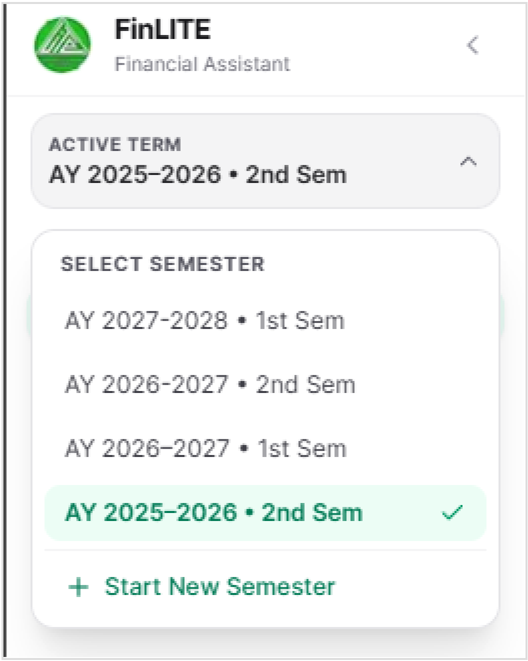
Financial Ledger & Transactions

Real-time auditable record of organization inflows and disbursements

Transaction & Event	Date	Category	Amount	RECORDED BY	RECEIPTS & ACTIONS
213 Revision Needed	Oct 20, 2026	Supplies & Materials	-₱100.00	Andrei John Geron... ADMIN	— 2
432	Oct 14, 2026	Tournament & E-Sports Registration Fees	+₱343.00	Andrei John Geron... ADMIN	—
Declared Cash Shortage (Adviser Approved) Cash Count Audit	Oct 14, 2026	Declared Cash Shortage Discrepancy	-₱20,317,126.92	Andrei John Geron... ADMIN	— 1
lanyarddd	Oct 13, 2026	Tournament & E-Sports Registration Fees	-₱4,567.00	System	—
none 2026	Oct 13, 2026	Tournament & E-Sports Registration Fees	+₱10.00	Andrei John Geron... ADMIN	—
gfghgjghjghj	Oct 13, 2026	Tournament & E-Sports Registration Fees	+₱10,000,999.00	System	—
gfghgjghjghj	Oct 13, 2026	Tournament & E-Sports Registration Fees	+₱10,000,999.00	System	—
T-shirt Club Week 2077 Settled · Boss G	Oct 11, 2026	Supplies & Materials	-₱150.00	Prohacker Weaver ADMIN	1 Receipt

Figure 16. FinLITE Executive Dashboard

This figure shows the main executive dashboard of the FinLITE system, serving as the central operational hub for monitoring organizational liquidity and daily financial activities. It features quick-action header buttons to record transactions, launch the cashbox audit keypad, compile liquidation reports, or open the AI Co-Pilot, accompanied by six top-level financial summary cards displaying Cashbox Physical Cash, Net Position, Student Receivables, Total Inflows, Total Outflows, and Personal Advances (Abono), alongside a real-time searchable Financial Ledger table with category filters and audit status indicators. Also here, the officer can monitor overall treasury reserves, filter transactions across ledger categories, switch between spreadsheet and card views, and inspect recent financial entries.

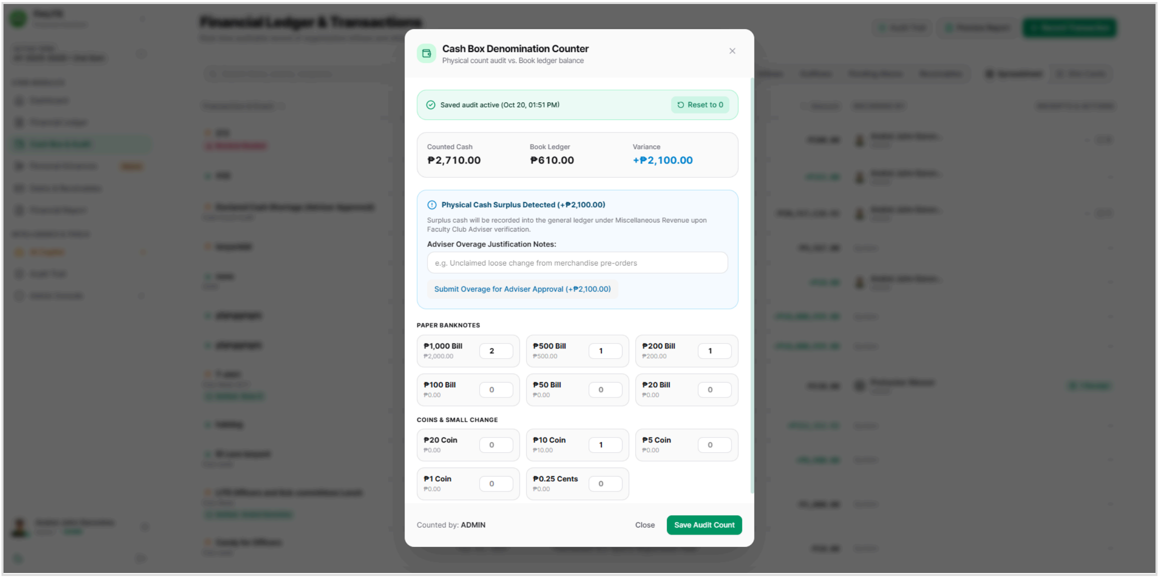


FinLITE
Financial Assistant
SELECT SEMESTER
AY 2027-2028 • 1st Sem
AY 2026-2027 • 2nd Sem
AY 2026–2027 • 1st Sem
AY 2025–2026 • 2nd Sem ✓
+ Start New Semester

ACTIVE TERM
AY 2025–2026 • 2nd Sem

Figure 17. Academic Term & Semester Selector

This figure shows the academic term and semester selection dropdown popover integrated into the application sidebar. It displays a chronological list of academic periods, including AY 2027–2028 1st Semester, AY 2026–2027 2nd Semester, AY 2026–2027 1st Semester, and the currently active AY 2025–2026 2nd Semester, alongside a "+ Start New Semester" action button. Also here, the officer can switch the active working semester to partition financial transactions, reconciliations, and debts, or initialize a new academic tenure with carried-over surplus balances.



Cash Box Denomination Counter
Physical count audit vs. Book ledger balance

Saved audit active (Oct 20, 01:51 PM)

Counted Cash	Book Ledger	Variance
₱2,710.00	₱610.00	+₱2,100.00

Physical Cash Surplus Detected (+₱2,100.00)

Surplus cash will be recorded into the general ledger under Miscellaneous Revenue upon Faculty Club Adviser verification.

Adviser Overage Justification Notes:

e.g. Unclaimed loose change from merchandise pre-orders

PAPER BANKNOTES

₱1,000 Bill ₱2,000.00	2	
₱500 Bill ₱500.00	1	
₱200 Bill ₱200.00	1	
₱100 Bill ₱0.00	0	
₱50 Bill ₱0.00	0	
₱20 Bill ₱0.00	0	

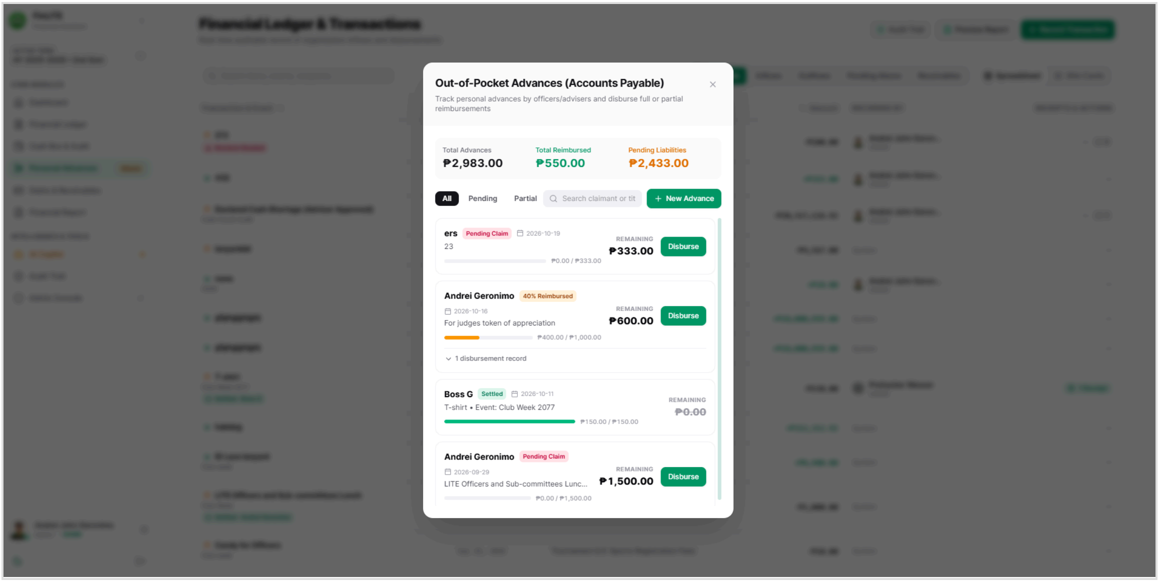
COINS & SMALL CHANGE

₱20 Coin ₱0.00	0	
₱10 Coin ₱10.00	1	
₱5 Coin ₱0.00	0	
₱1 Coin ₱0.00	0	
₱0.25 Cents ₱0.00	0	

Counted by: ADMIN

Figure 18. Cash Box Denomination Counter & Physical Reconciliation

This figure shows the physical cash denomination counter and variance audit modal used during dual-officer cashbox reconciliations. It displays the active session timestamp, live calculated metrics for Counted Cash, Book Ledger balance, and computed Variance, an automated surplus/shortage alert card with an Adviser Justification input field, and an interactive digital keypad covering all 11 legal tender Philippine currency units divided into Paper Banknotes (₱1,000, ₱500, ₱200, ₱100, ₱50, ₱20) and Coins & Small Change (₱20, ₱10, ₱5, ₱1, ₱0.25 Cents). Also here, the Treasurer and Auditor can enter physical piece-counts for each currency denomination, inspect real-time variance calculations, draft cash shortage justifications for adviser sign-off, and commit verified audit counts to the database.



Out-of-Pocket Advances (Accounts Payable)

Track personal advances by officers/advisers and disburse full or partial reimbursements

Total Advances ₱2,983.00
Total Reimbursed ₱550.00
Pending Liabilities ₱2,433.00

ers Pending Claim 2026-10-19
23
₱0.00 / ₱333.00
REMAINING ₱333.00

Andrei Geronimo 40% Reimbursed
2026-10-16
For judges token of appreciation
₱400.00 / ₱1,000.00
REMAINING ₱600.00

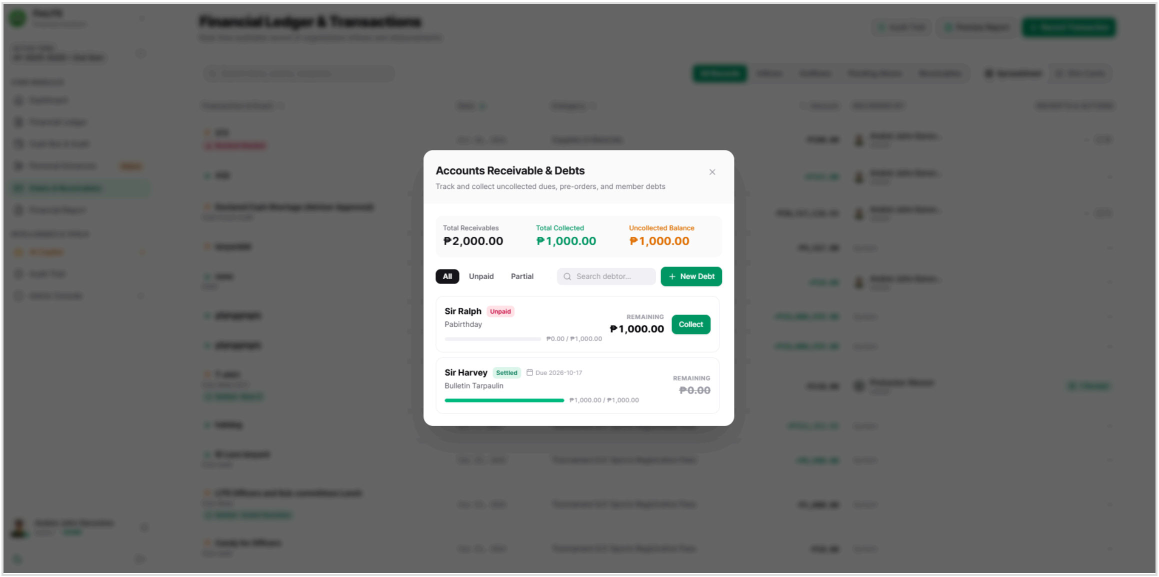
1 disbursement record

Boss G Settled 2026-10-11
T-shirt • Event: Club Week 2077
₱150.00 / ₱150.00
REMAINING ~~₱0.00~~

Andrei Geronimo Pending Claim
2026-09-29
LITE Officers and Sub-committees Lunc...
₱0.00 / ₱1,500.00
REMAINING ₱1,500.00

Figure 19. Out-of-Pocket Advances (Abono) Queue

This figure shows the Out-of-Pocket Advances (Accounts Payable) management modal designed to regulate personal expenditures incurred by officers and advisers for official club activities. It features summary metric cards for Total Advances, Total Reimbursed, and Pending Liabilities, alongside status filter tabs (All, Pending, Partial), a search bar, a "+ New Advance" button, and itemized advance cards displaying claimant names, claim status badges, submission dates, expense particulars, visual progress bars of reimbursed amounts, and "Disburse" payout action buttons. Also here, the officer can track outstanding club liabilities, file new advance claims with attached receipt images, monitor repayment progress, and disburse cashbox reimbursements.



Accounts Receivable & Debts

Track and collect uncollected dues, pre-orders, and member debts

Total Receivables ₱2,000.00

Total Collected ₱1,000.00

Uncollected Balance ₱1,000.00

Sir Ralph Unpaid

Pabirthday

₱0.00 / ₱1,000.00

REMAINING ₱1,000.00

Sir Harvey Settled Due 2026-10-17

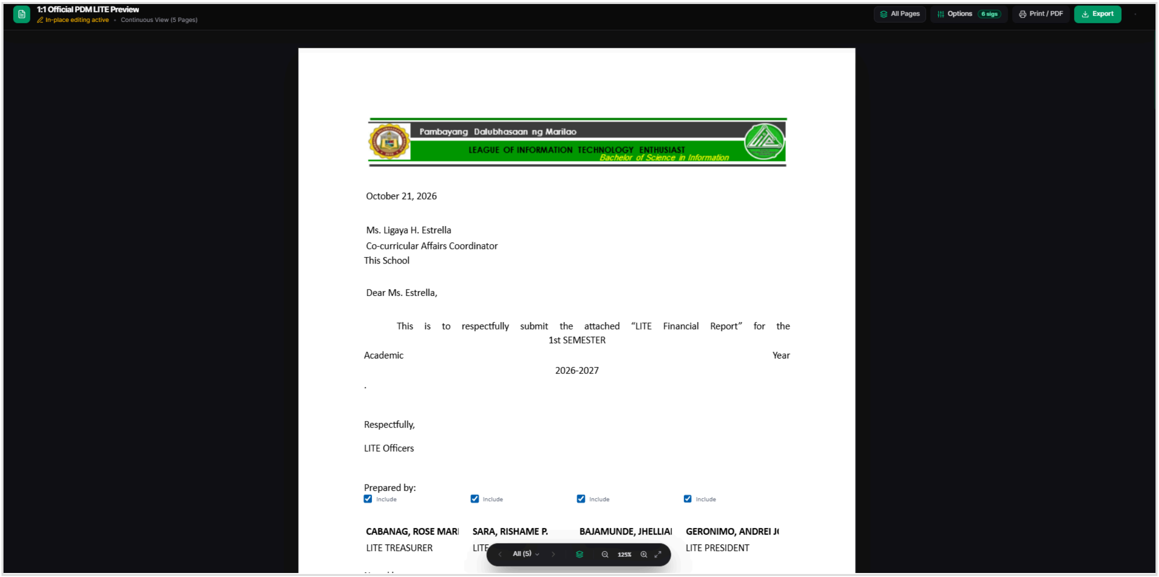
Bulletin Tarpaulin

₱1,000.00 / ₱1,000.00

REMAINING ~~₱0.00~~

Figure 20. Accounts Receivable & Debts Subsystem

This figure shows the Accounts Receivable & Debts tracking modal used to manage uncollected student merchandise pre-orders and tournament entry fees. It displays summary metrics for Total Receivables, Total Collected, and Uncollected Balance, along with filter tabs (All, Unpaid, Partial), a debtor search input, a "+ New Debt" button, and detailed receivable cards indicating student debtor names, payment status badges, purpose descriptions, remaining balances, and a "Collect" payment recording button. Also here, the Treasurer can log student installment pre-orders, search debtor accounts, track partial payments, and record cash collections directly into the ledger.


1:1 Official PDM LITE Preview
In-place editing active · Continuous View (5 Pages)
All Pages
Options 6 sigs
Print / PDF
Export
Pambayang Dalubhasaan ng Marilao
LEAGUE OF INFORMATION TECHNOLOGY ENTHUSIAST
Bachelor of Science in Information
All (5)
125%

October 21, 2026

Ms. Ligaya H. Estrella
Co-curricular Affairs Coordinator
This School

Dear Ms. Estrella,

This is to respectfully submit the attached "LITE Financial Report" for the 1st SEMESTER Academic Year 2026-2027
.

Respectfully,

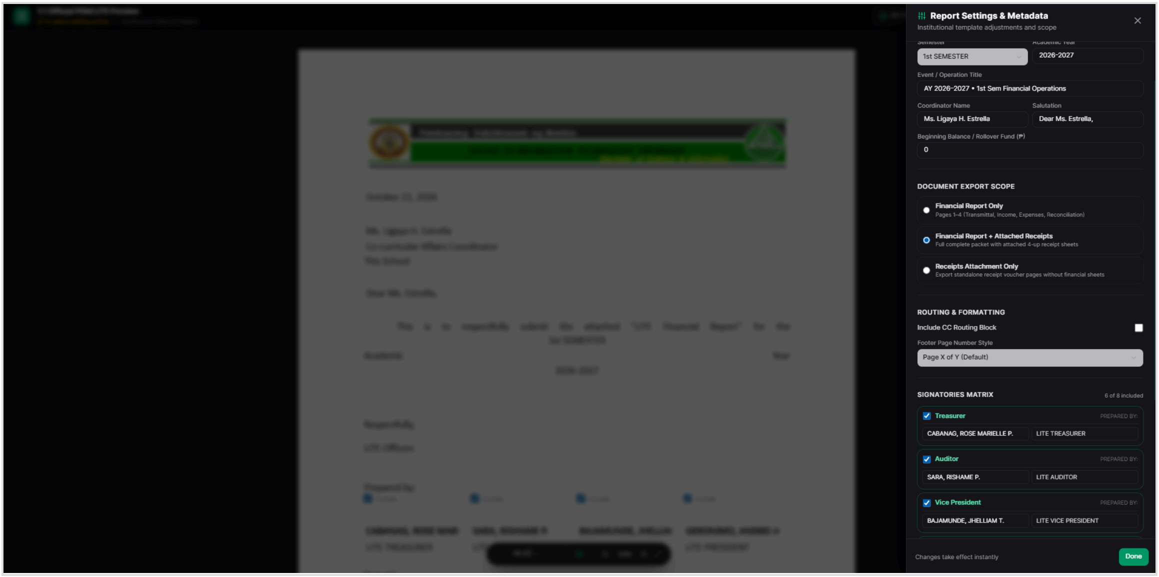
LITE Officers

Prepared by:

☑ Include	☑ Include	☑ Include	☑ Include
CABANAG, ROSE MARI	**SARA, RISHAME P.**	**BAJAMUNDE, JHELLIAI**	**GERONIMO, ANDREI J**
LITE TREASURER	LITE		LITE PRESIDENT

Figure 21. 1:1 Official PDM LITE Institutional Report Preview

This figure shows the 1:1 pixel-accurate live preview of the formal institutional financial liquidation report in the browser. It renders the official PDM and LITE header letterhead banner, transmittal date, formal addressee block to Co-curricular Affairs Coordinator Ms. Ligaya H. Estrella, respectful submission statement, and the preliminary officer signatory block (Treasurer, Auditor, President) with individual "Include" toggle checkboxes. Also here, the officer can inspect the exact physical page layout of the liquidation packet, make in-place text edits, toggle required administrative signatures, and trigger export to Microsoft Word (.docx) or PDF.



Report Settings & Metadata
Institutional template adjustments and scope

Semester: 1st SEMESTER
Academic Year: 2026-2027
Event / Operation Title: AY 2026–2027 • 1st Sem Financial Operations
Coordinator Name: Ms. Ligaya H. Estrella
Salutation: Dear Ms. Estrella,
Beginning Balance / Rollover Fund (₱): 0

DOCUMENT EXPORT SCOPE

○ Financial Report Only
Pages 1–4 (Transmittal, Income, Expenses, Reconciliation)

● Financial Report + Attached Receipts
Full complete packet with attached 4-up receipt sheets

○ Receipts Attachment Only
Export standalone receipt voucher pages without financial sheets

ROUTING & FORMATTING

Include CC Routing Block ☐
Footer Page Number Style: Page X of Y (Default)

SIGNATORIES MATRIX — 6 of 8 included

☑ Treasurer — PREPARED BY:
CABANAG, ROSE MARIELLE P. — LITE TREASURER

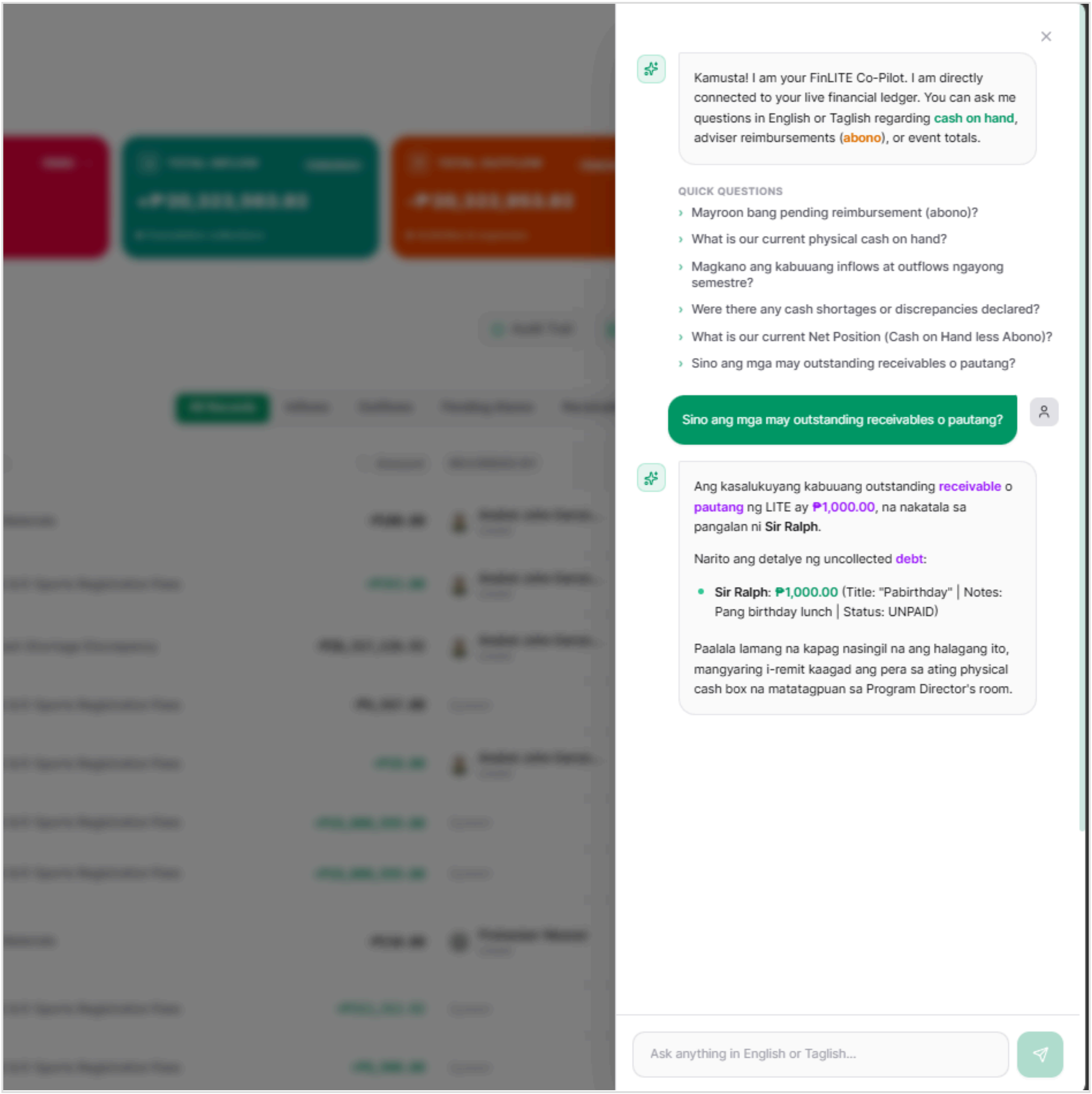
☑ Auditor — PREPARED BY:
SARA, RISHAME P. — LITE AUDITOR

☑ Vice President — PREPARED BY:
BAJAMUNDE, JHELLIAM T. — LITE VICE PRESIDENT

Changes take effect instantly

Figure 22. Report Settings & Metadata Configuration Drawer

This figure shows the slide-out Report Settings and Metadata configuration drawer for institutional liquidation reports. It provides input fields for Semester, Academic Year, Event Title, Coordinator Name, Salutation, and Beginning Balance, alongside radio selectors for Document Export Scope (Financial Report Only, Financial Report + Attached Receipts, Receipts Attachment Only), routing checkboxes, footer page numbering styles, and an interactive Signatories Matrix with checkboxes to include or exclude specific officers. Also here, the officer can adjust institutional transmittal metadata, select document compilation scopes, customize routing blocks, and save template preferences before generating reports.



Kamusta! I am your FinLITE Co-Pilot. I am directly connected to your live financial ledger. You can ask me questions in English or Taglish regarding cash on hand, adviser reimbursements (abono), or event totals.

QUICK QUESTIONS

- Mayroon bang pending reimbursement (abono)?
- What is our current physical cash on hand?
- Magkano ang kabuuang inflows at outflows ngayong semestre?
- Were there any cash shortages or discrepancies declared?
- What is our current Net Position (Cash on Hand less Abono)?
- Sino ang mga may outstanding receivables o pautang?

Sino ang mga may outstanding receivables o pautang?

Ang kasalukuyang kabuuang outstanding receivable o pautang ng LITE ay ₱1,000.00, na nakatala sa pangalan ni **Sir Ralph**.

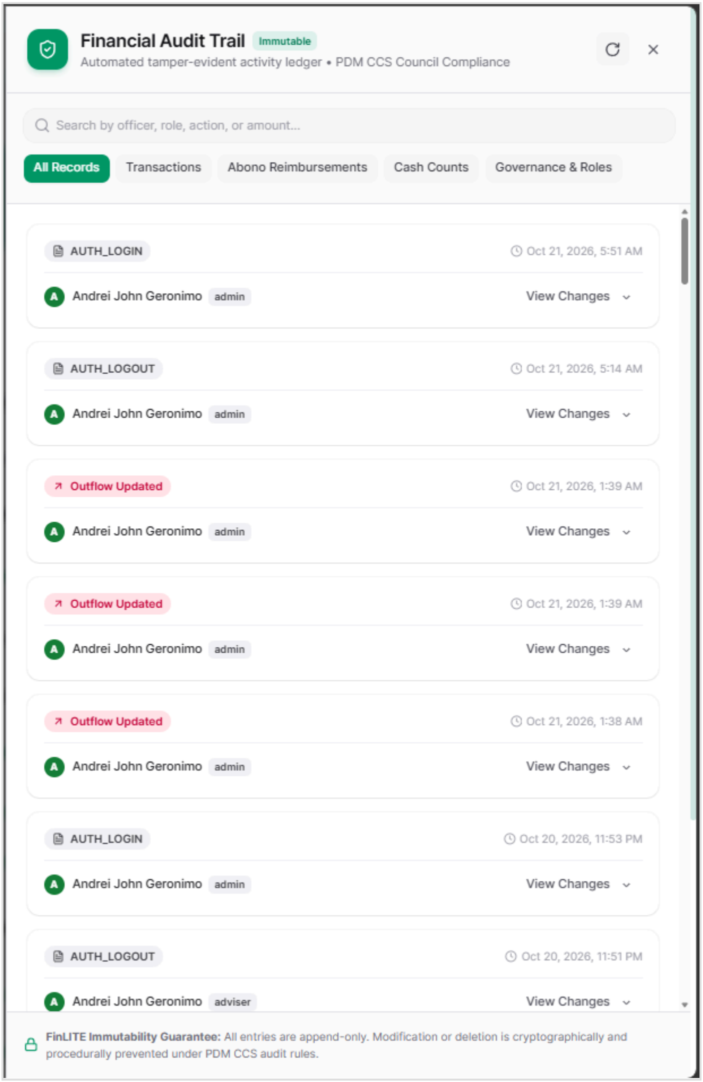
Narito ang detalye ng uncollected debt:

- **Sir Ralph**: ₱1,000.00 (Title: "Pabirthday" | Notes: Pang birthday lunch | Status: UNPAID)

Paalala lamang na kapag nasingil na ang halagang ito, mangyaring i-remit kaagad ang pera sa ating physical cash box na matatagpuan sa Program Director's room.

Figure 23. Grounded AI Financial Co-Pilot Conversational Drawer

This figure shows the conversational AI Financial Co-Pilot drawer interface, accessible across all officer dashboards. It displays a bilingual welcome greeting, clickable Quick Questions prompt chips (such as checking pending reimbursements or cash on hand), a conversational thread featuring user queries in Taglish ("Sino ang mga may outstanding receivables o pautang?"), and an AI response containing exact debtor names, amounts, notes, and custodial cashbox reminders grounded directly in verified database records, along with a prompt input bar. Also here, the officer can submit natural-language financial inquiries in Taglish or English, inspect verified budget summaries, and receive instant, zero-hallucination treasury insights.


Financial Audit Trail
Immutable
Automated tamper-evident activity ledger • PDM CCS Council Compliance
Search by officer, role, action, or amount...
All Records
Transactions
Abono Reimbursements
Cash Counts
Governance & Roles

AUTH_LOGIN Oct 21, 2026, 5:51 AM
Andrei John Geronimo admin View Changes

AUTH_LOGOUT Oct 21, 2026, 5:14 AM
Andrei John Geronimo admin View Changes

Outflow Updated Oct 21, 2026, 1:39 AM
Andrei John Geronimo admin View Changes

Outflow Updated Oct 21, 2026, 1:39 AM
Andrei John Geronimo admin View Changes

Outflow Updated Oct 21, 2026, 1:38 AM
Andrei John Geronimo admin View Changes

AUTH_LOGIN Oct 20, 2026, 11:53 PM
Andrei John Geronimo admin View Changes

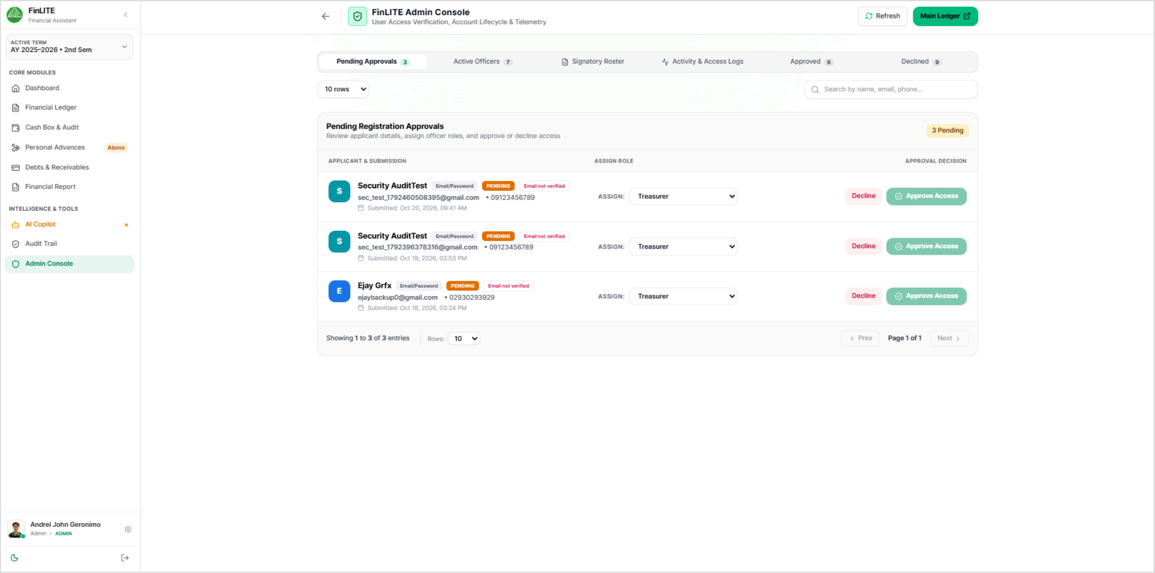
AUTH_LOGOUT Oct 20, 2026, 11:51 PM
Andrei John Geronimo adviser View Changes

FinLITE Immutability Guarantee: All entries are append-only. Modification or deletion is cryptographically and procedurally prevented under PDM CCS audit rules.

Figure 24. Financial Audit Trail & Tamper-Evident Ledger

This figure shows the Financial Audit Trail sliding drawer, providing an immutable record of system actions in compliance with PDM CCS student council audit standards. It displays a search bar, category filter pills (All Records, Transactions, Abono Reimbursements, Cash Counts, Governance & Roles), and an append-only chronological feed showing action badges (such as AUTH_LOGIN, AUTH_LOGOUT, and Outflow

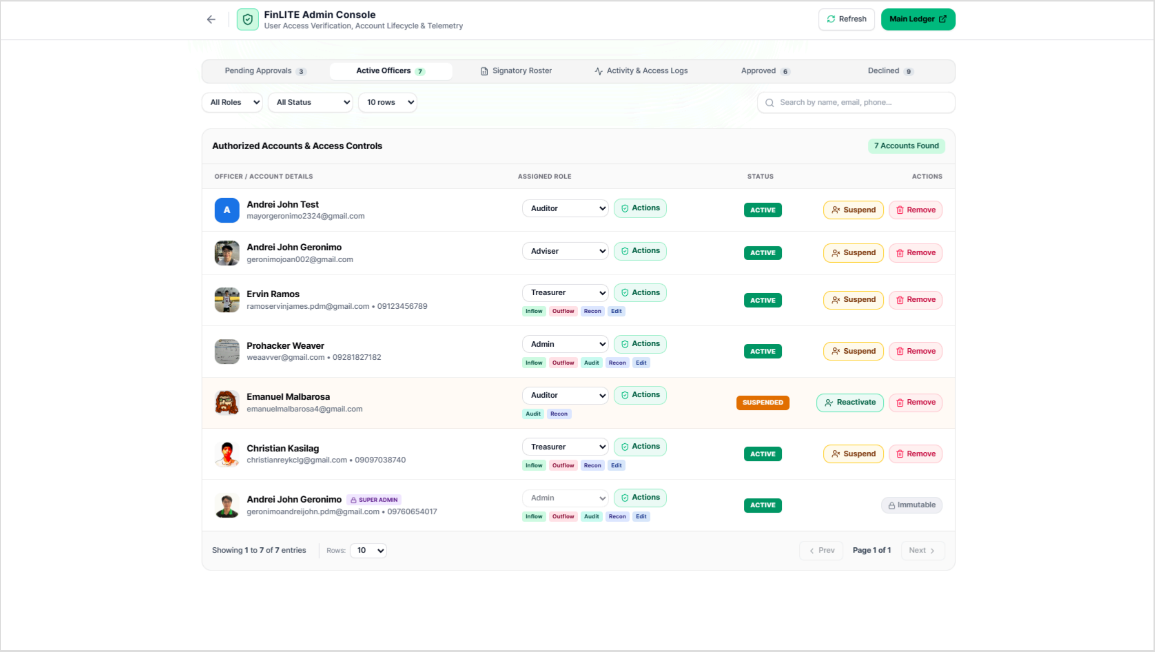
Updated), timestamps, acting officer profile badges, and collapsible "View Changes" diff viewers, anchored by an immutability guarantee notice. Also here, the Auditor and Club Advisers can inspect chronological system modifications, search events by user or action type, and audit historical record changes.


FinLITE
Financial Assistant
ACTIVE TERM
AY 2025–2026 • 2nd Sem
CORE MODULES
Dashboard
Financial Ledger
Cash Box & Audit
Personal Advances
Abono
Debts & Receivables
Financial Report
INTELLIGENCE & TOOLS
AI Copilot
Audit Trail
Admin Console
Andrei John Geronimo
Admin • ADMIN
FinLITE Admin Console
User Access Verification, Account Lifecycle & Telemetry
Refresh
Main Ledger
Pending Approvals 3
Active Officers 7
Signatory Roster
Activity & Access Logs
Approved 6
Declined 9
10 rows
Search by name, email, phone...
Pending Registration Approvals
Review applicant details, assign officer roles, and approve or decline access
3 Pending
Showing 1 to 3 of 3 entries
Rows: 10
Prev
Page 1 of 1
Next

APPLICANT & SUBMISSION	ASSIGN ROLE	APPROVAL DECISION
Security AuditTest Email/Password PENDING Email not verified sec_test_1792460508395@gmail.com • 09123456789 Submitted: Oct 20, 2026, 09:41 AM	ASSIGN: Treasurer	Decline Approve Access
Security AuditTest Email/Password PENDING Email not verified sec_test_1792396378316@gmail.com • 09123456789 Submitted: Oct 19, 2026, 03:53 PM	ASSIGN: Treasurer	Decline Approve Access
Ejay Grfx Email/Password PENDING Email not verified ejaybackup0@gmail.com • 02930293929 Submitted: Oct 18, 2026, 03:24 PM	ASSIGN: Treasurer	Decline Approve Access

Figure 25. Admin Console: Pending Registration Approvals

This figure shows the Pending Approvals tab within the FinLITE Admin Console (/admin), used for reviewing new applicant registrations. It features summary navigation tabs (Pending Approvals, Active Officers, Signatory Roster, Activity & Access Logs, Approved, Declined) and an approval queue displaying applicant cards with submission timestamps, emails, contact numbers, authentication provider badges, assigned role dropdowns, and "Decline" / "Approve Access" decision buttons. Also here, the Club Adviser can review pending officer registrations, designate approved roles, reject invalid accounts with written justification, and grant platform access.


FinLITE Admin Console
User Access Verification, Account Lifecycle & Telemetry
Refresh
Main Ledger
Pending Approvals 3
Active Officers 7
Signatory Roster
Activity & Access Logs
Approved 6
Declined 9
All Roles
All Status
10 rows
Search by name, email, phone...

Authorized Accounts & Access Controls

7 Accounts Found

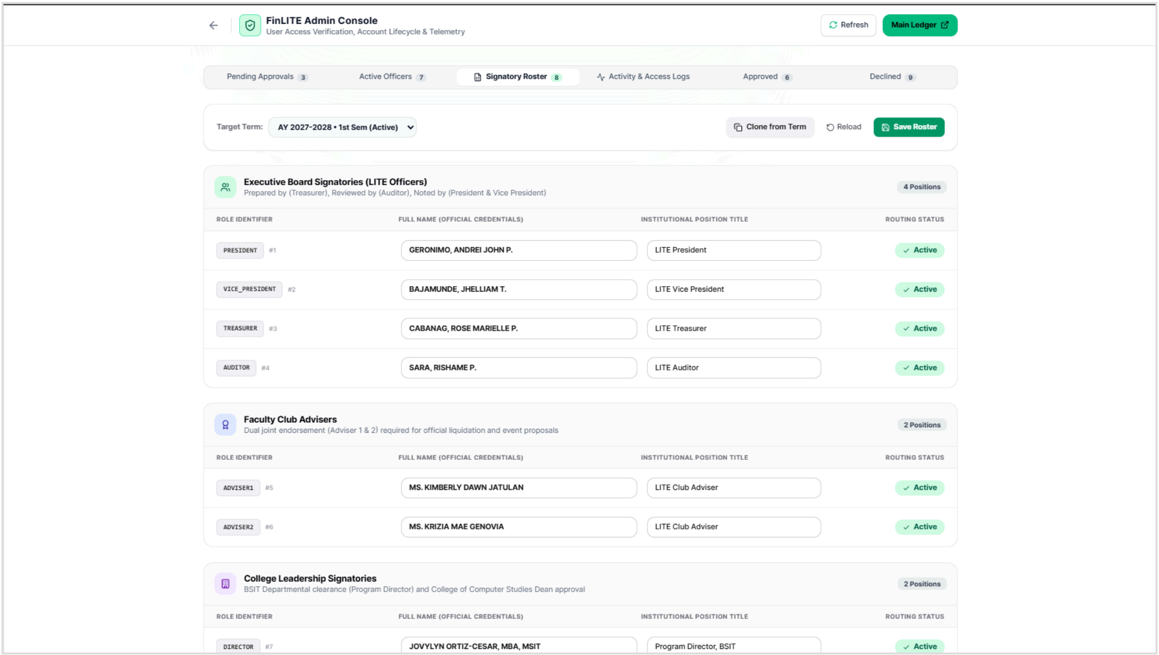
OFFICER / ACCOUNT DETAILS	ASSIGNED ROLE	STATUS	ACTIONS
Andrei John Test mayorgeronimo2324@gmail.com	Auditor · Actions	ACTIVE	Suspend · Remove
Andrei John Geronimo geronimojoan002@gmail.com	Adviser · Actions	ACTIVE	Suspend · Remove
Ervin Ramos ramoservinjames.pdm@gmail.com • 09123456789	Treasurer · Actions Inflow Outflow Recon Edit	ACTIVE	Suspend · Remove
Prohacker Weaver weaavver@gmail.com • 09281827182	Admin · Actions Inflow Outflow Audit Recon Edit	ACTIVE	Suspend · Remove
Emanuel Malbarosa emanuelmalbarosa4@gmail.com	Auditor · Actions Audit Recon	SUSPENDED	Reactivate · Remove
Christian Kasilag christianreykclg@gmail.com • 09097038740	Treasurer · Actions Inflow Outflow Recon Edit	ACTIVE	Suspend · Remove
Andrei John Geronimo SUPER ADMIN geronimoandreijohn.pdm@gmail.com • 09760654017	Admin · Actions Inflow Outflow Audit Recon Edit	ACTIVE	Immutable

Showing 1 to 7 of 7 entries

Rows: 10
Prev
Page 1 of 1
Next

Figure 26. Admin Console: Active Officers & Access Controls

This figure shows the Active Officers tab of the Admin Console, displaying the roster of authorized user accounts. It presents officer profile avatars, full names, email addresses, contact numbers, Super Admin badges, assigned role dropdowns, granular permission capability pills (Inflow, Outflow, Audit, Recon, Edit), account status badges (Active, Suspended), "Actions" modals, "Suspend" / "Reactivate" buttons, and an immutable lock indicator on the primary master administrator account. Also here, the Club Adviser can reassign officer roles, review custom capability tags, suspend or reactivate accounts, and manage user permissions.


FinLITE Admin Console
User Access Verification, Account Lifecycle & Telemetry
Refresh
Main Ledger
Pending Approvals 3
Active Officers 7
Signatory Roster 8
Activity & Access Logs
Approved 6
Declined 9
Clone from Term
Reload
Save Roster

Target Term: AY 2027-2028 • 1st Sem (Active)

Executive Board Signatories (LITE Officers)
Prepared by (Treasurer), Reviewed by (Auditor), Noted by (President & Vice President)
4 Positions

ROLE IDENTIFIER	FULL NAME (OFFICIAL CREDENTIALS)	INSTITUTIONAL POSITION TITLE	ROUTING STATUS
`PRESIDENT` #1	GERONIMO, ANDREI JOHN P.	LITE President	✓ Active
`VICE_PRESIDENT` #2	BAJAMUNDE, JHELLIAM T.	LITE Vice President	✓ Active
`TREASURER` #3	CABANAG, ROSE MARIELLE P.	LITE Treasurer	✓ Active
`AUDITOR` #4	SARA, RISHAME P.	LITE Auditor	✓ Active

Faculty Club Advisers
Dual joint endorsement (Adviser 1 & 2) required for official liquidation and event proposals
2 Positions

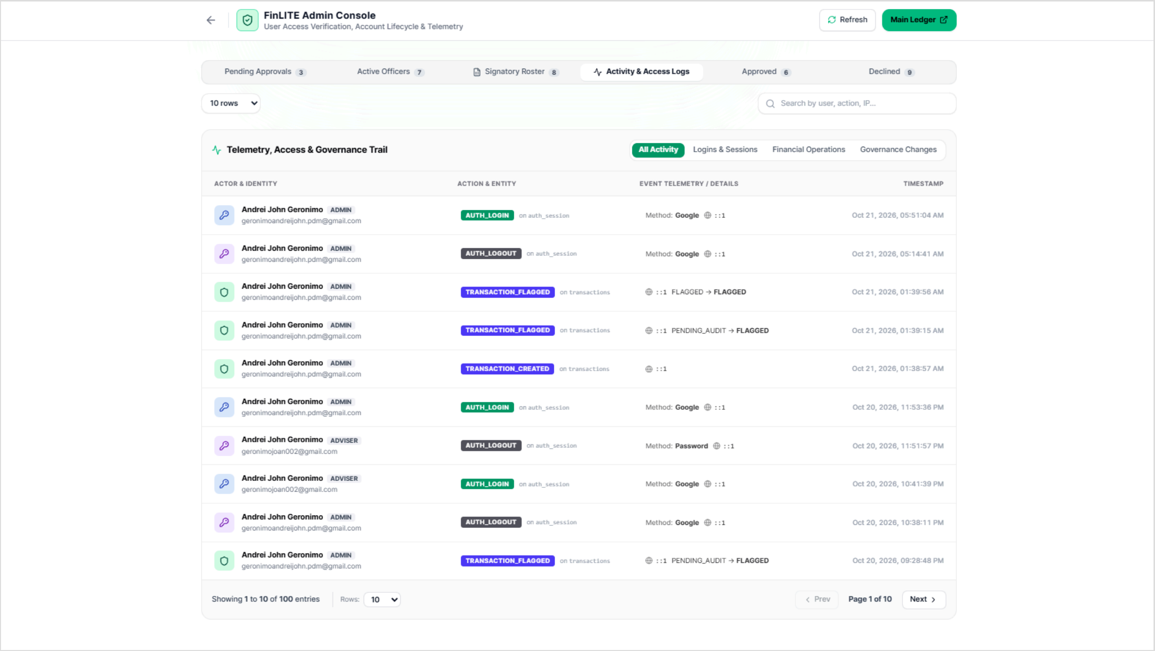
ROLE IDENTIFIER	FULL NAME (OFFICIAL CREDENTIALS)	INSTITUTIONAL POSITION TITLE	ROUTING STATUS
`ADVISER1` #5	MS. KIMBERLY DAWN JATULAN	LITE Club Adviser	✓ Active
`ADVISER2` #6	MS. KRIZIA MAE GENOVIA	LITE Club Adviser	✓ Active

College Leadership Signatories
BSIT Departmental clearance (Program Director) and College of Computer Studies Dean approval
2 Positions

ROLE IDENTIFIER	FULL NAME (OFFICIAL CREDENTIALS)	INSTITUTIONAL POSITION TITLE	ROUTING STATUS
`DIRECTOR` #7	JOVYLYN ORTIZ-CESAR, MBA, MSIT	Program Director, BSIT	✓ Active

Figure 27. Admin Console: Dynamic Signatory Roster

This figure shows the Signatory Roster configuration interface within the Admin Console. It features a target academic term selector, "Clone from Term" button, "Save Roster" button, and structured administrative tiers including Executive Board Signatories (President, Vice President, Treasurer, Auditor), Faculty Club Advisers, and College Leadership Signatories, complete with editable input fields for official full names with credentials, institutional position titles, and routing statuses. Also here, the Club Adviser can update official signatory names and credentials, clone rosters across semesters, and save routing configurations for automated report generation.


FinLITE Admin Console
User Access Verification, Account Lifecycle & Telemetry
Refresh
Main Ledger
Pending Approvals 3
Active Officers 7
Signatory Roster 8
Activity & Access Logs
Approved 6
Declined 9
10 rows
Search by user, action, IP...
Telemetry, Access & Governance Trail
All Activity
Logins & Sessions
Financial Operations
Governance Changes

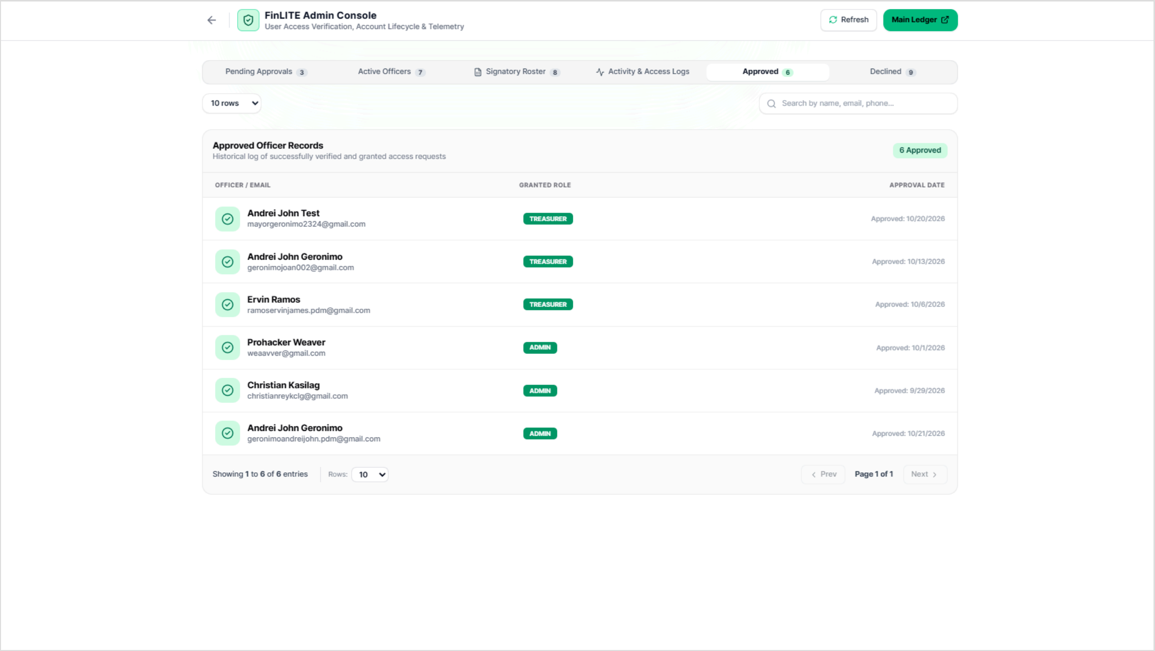
ACTOR & IDENTITY	ACTION & ENTITY	EVENT TELEMETRY / DETAILS	TIMESTAMP
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	AUTH_LOGIN on auth_session	Method: Google ::1	Oct 21, 2026, 05:51:04 AM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	AUTH_LOGOUT on auth_session	Method: Google ::1	Oct 21, 2026, 05:14:41 AM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	TRANSACTION_FLAGGED on transactions	::1 FLAGGED → FLAGGED	Oct 21, 2026, 01:39:56 AM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	TRANSACTION_FLAGGED on transactions	::1 PENDING_AUDIT → FLAGGED	Oct 21, 2026, 01:39:15 AM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	TRANSACTION_CREATED on transactions	::1	Oct 21, 2026, 01:38:57 AM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	AUTH_LOGIN on auth_session	Method: Google ::1	Oct 20, 2026, 11:53:36 PM
Andrei John Geronimo ADVISER geronimojoan002@gmail.com	AUTH_LOGOUT on auth_session	Method: Password ::1	Oct 20, 2026, 11:51:57 PM
Andrei John Geronimo ADVISER geronimojoan002@gmail.com	AUTH_LOGIN on auth_session	Method: Google ::1	Oct 20, 2026, 10:41:39 PM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	AUTH_LOGOUT on auth_session	Method: Google ::1	Oct 20, 2026, 10:38:11 PM
Andrei John Geronimo ADMIN geronimoandreijohn.pdm@gmail.com	TRANSACTION_FLAGGED on transactions	::1 PENDING_AUDIT → FLAGGED	Oct 20, 2026, 09:28:48 PM

Showing 1 to 10 of 100 entries

Rows: 10
Prev
Page 1 of 10
Next

Figure 28. Admin Console: Telemetry, Access & Governance Trail

This figure shows the Telemetry, Access & Governance Trail tab of the Admin Console, maintaining a complete log of administrative and authentication events. It displays filter pills (All Activity, Logins & Sessions, Financial Operations, Governance Changes), a search box, and chronological log rows detailing actor identity and email, role badges, action and entity tags (such as AUTH_LOGIN, AUTH_LOGOUT, TRANSACTION_FLAGGED, and TRANSACTION_CREATED), event telemetry details (method, IP address, state transitions), and exact timestamps. Also here, the Club Adviser can audit system access events, monitor session authentication history, track financial record modifications, and verify administrative security trails.



FinLITE Admin Console
User Access Verification, Account Lifecycle & Telemetry
Refresh
Main Ledger
Pending Approvals 3
Active Officers 7
Signatory Roster 8
Activity & Access Logs
Approved 6
Declined 9
10 rows
Search by name, email, phone...

Approved Officer Records
Historical log of successfully verified and granted access requests

6 Approved

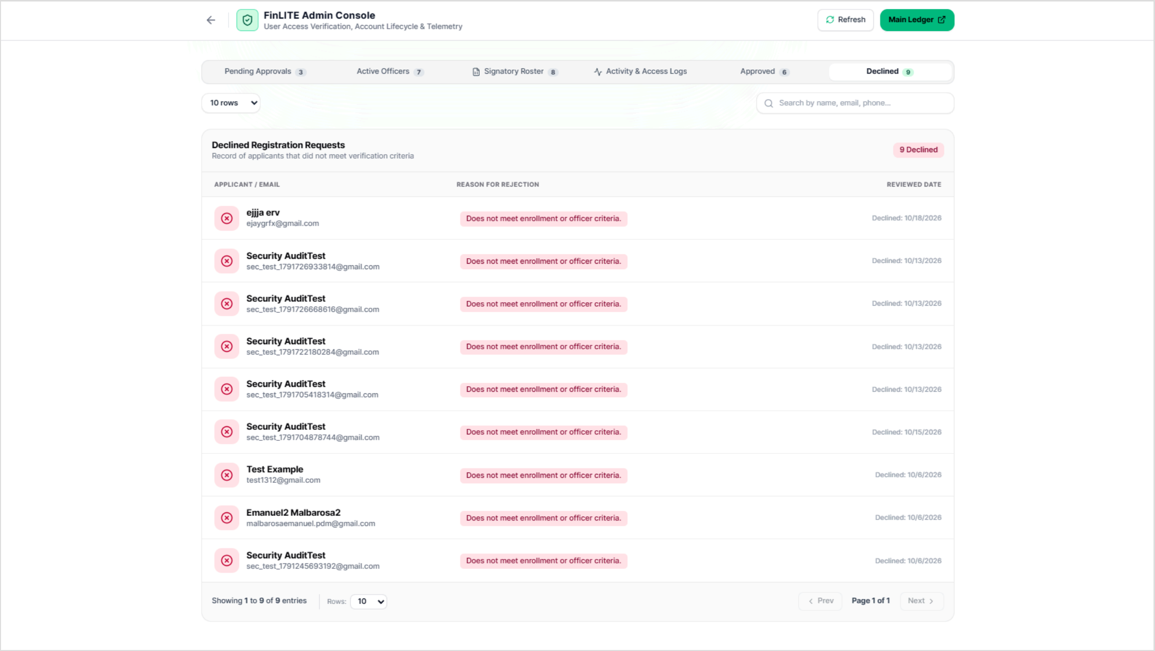
OFFICER / EMAIL	GRANTED ROLE	APPROVAL DATE
✓ **Andrei John Test** mayorgeronimo2324@gmail.com	TREASURER	Approved: 10/20/2026
✓ **Andrei John Geronimo** geronimojoan002@gmail.com	TREASURER	Approved: 10/13/2026
✓ **Ervin Ramos** ramoservinjames.pdm@gmail.com	TREASURER	Approved: 10/6/2026
✓ **Prohacker Weaver** weaavver@gmail.com	ADMIN	Approved: 10/1/2026
✓ **Christian Kasilag** christianreykclg@gmail.com	ADMIN	Approved: 9/29/2026
✓ **Andrei John Geronimo** geronimoandreijohn.pdm@gmail.com	ADMIN	Approved: 10/21/2026

Showing 1 to 6 of 6 entries

Rows: 10
Prev
Page 1 of 1
Next

Figure 29. Admin Console: Approved Officer Records

This figure shows the Approved Officer Records tab in the Admin Console, preserving a permanent historical log of verified access requests. It displays verified checkmark badges, approved applicant names, email addresses, granted role badges (Treasurer, Admin), and the exact calendar dates when account approvals were granted. Also here, the Club Adviser can review the historical archive of successfully approved officer accounts and verify past onboarding records.


FinLITE Admin Console
User Access Verification, Account Lifecycle & Telemetry
Refresh
Main Ledger
Pending Approvals 3
Active Officers 7
Signatory Roster 8
Activity & Access Logs
Approved 6
Declined 9
10 rows
Search by name, email, phone...

Declined Registration Requests
Record of applicants that did not meet verification criteria

9 Declined

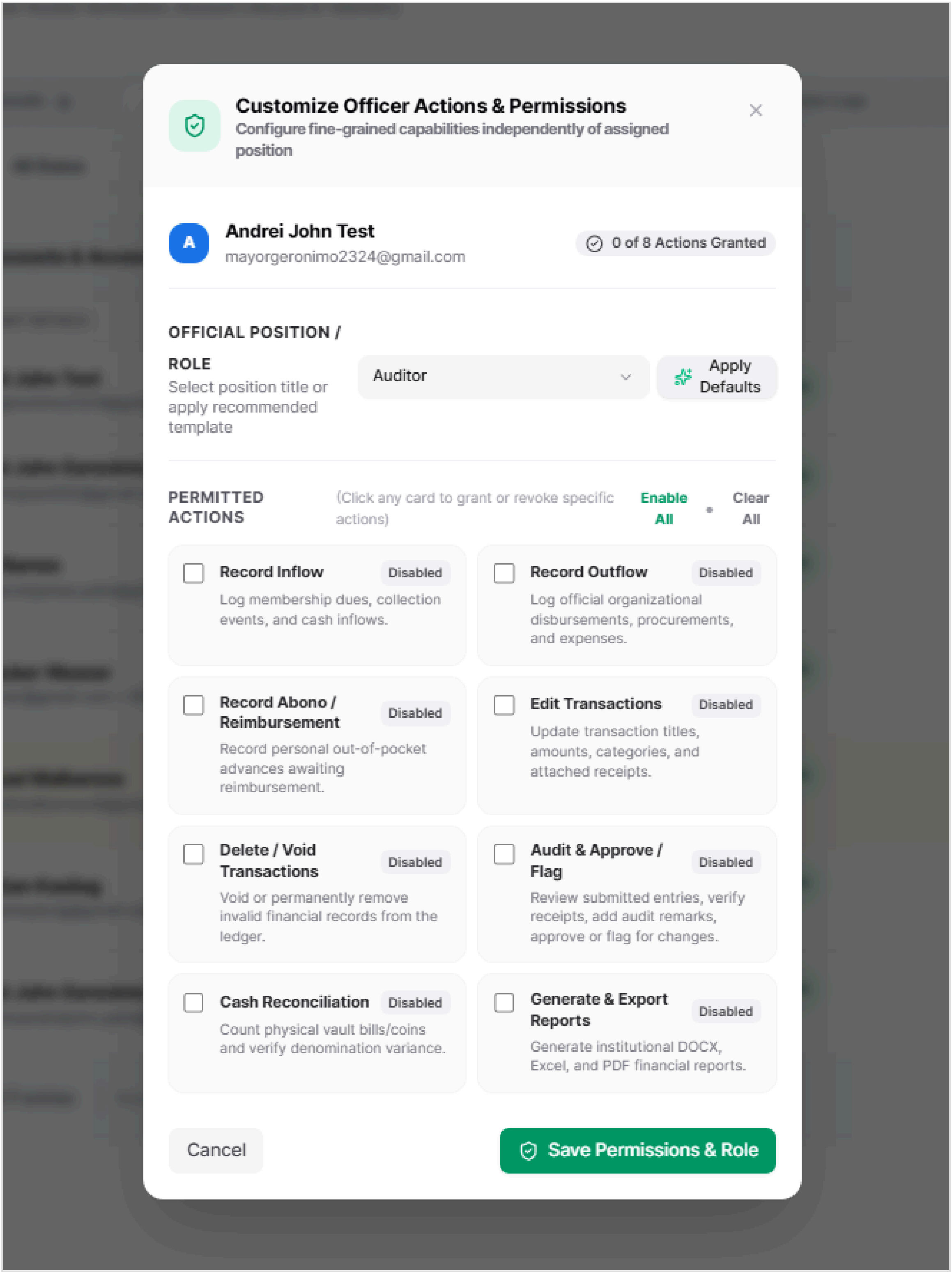
APPLICANT / EMAIL	REASON FOR REJECTION	REVIEWED DATE
ejjja erv ejaygrfx@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/18/2026
Security AuditTest sec_test_1791726933814@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/13/2026
Security AuditTest sec_test_1791726668616@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/13/2026
Security AuditTest sec_test_1791722180284@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/13/2026
Security AuditTest sec_test_1791705418314@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/13/2026
Security AuditTest sec_test_1791704878744@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/15/2026
Test Example test1312@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/6/2026
Emanuel2 Malbarosa2 malbarosaemanuel.pdm@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/6/2026
Security AuditTest sec_test_1791245693192@gmail.com	Does not meet enrollment or officer criteria.	Declined: 10/6/2026

Showing **1** to **9** of **9** entries

Rows: 10
Prev
Page 1 of 1
Next

Figure 30. Admin Console: Declined Registration Requests

This figure shows the Declined Registration Requests tab in the Admin Console, documenting applicant applications that failed to meet verification criteria. It displays declined status icons, applicant names and email addresses, recorded reasons for rejection (such as "Does not meet enrollment or officer criteria"), and review timestamps. Also here, the Club Adviser can review rejected applicant records and their recorded justifications.



Customize Officer Actions & Permissions
Configure fine-grained capabilities independently of assigned position

Andrei John Test
mayorgeronimo2324@gmail.com

0 of 8 Actions Granted

OFFICIAL POSITION / ROLE
Select position title or apply recommended template

Auditor

PERMITTED ACTIONS (Click any card to grant or revoke specific actions)

☐ **Record Inflow** Disabled
Log membership dues, collection events, and cash inflows.

☐ **Record Outflow** Disabled
Log official organizational disbursements, procurements, and expenses.

☐ **Record Abono / Reimbursement** Disabled
Record personal out-of-pocket advances awaiting reimbursement.

☐ **Edit Transactions** Disabled
Update transaction titles, amounts, categories, and attached receipts.

☐ **Delete / Void Transactions** Disabled
Void or permanently remove invalid financial records from the ledger.

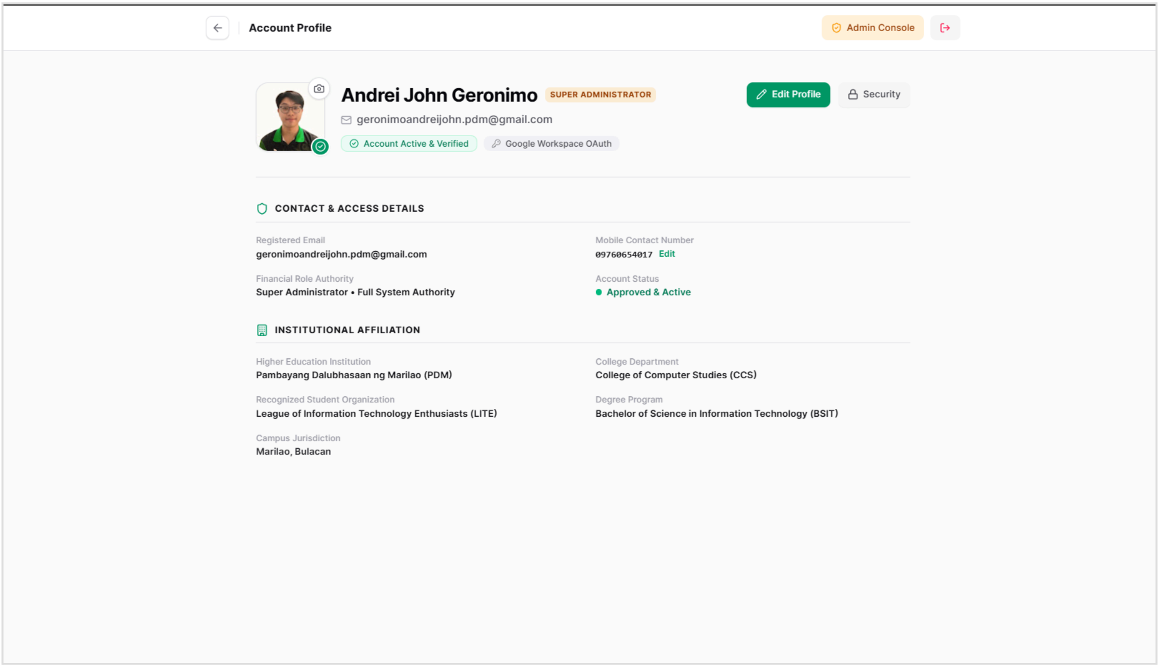
☐ **Audit & Approve / Flag** Disabled
Review submitted entries, verify receipts, add audit remarks, approve or flag for changes.

☐ **Cash Reconciliation** Disabled
Count physical vault bills/coins and verify denomination variance.

☐ **Generate & Export Reports** Disabled
Generate institutional DOCX, Excel, and PDF financial reports.

Figure 31. Customize Officer Actions & Permissions Modal

This figure shows the modal dialog for configuring fine-grained capabilities independently of assigned officer positions. It features the selected officer's avatar, name, and email, an Official Position/Role selector with an "Apply Defaults" button, and 8 granular capability checkbox cards (Record Inflow, Record Outflow, Record Abono/Reimbursement, Edit Transactions, Delete/Void Transactions, Audit & Approve/Flag, Cash Reconciliation, and Generate & Export Reports), alongside "Enable All" / "Clear All" shortcuts and a "Save Permissions & Role" button. Also here, the Club Adviser can dynamically grant, customize, or revoke operational capabilities for any officer via checkboxes, overriding default role boundaries during special events.



Andrei John Geronimo SUPER ADMINISTRATOR

geronimoandreijohn.pdm@gmail.com

Account Active & Verified
Google Workspace OAuth

CONTACT & ACCESS DETAILS

Registered Email
geronimoandreijohn.pdm@gmail.com

Mobile Contact Number
09760654017 Edit

Financial Role Authority
Super Administrator • Full System Authority

Account Status
● Approved & Active

INSTITUTIONAL AFFILIATION

Higher Education Institution
Pambayang Dalubhasaan ng Marilao (PDM)

College Department
College of Computer Studies (CCS)

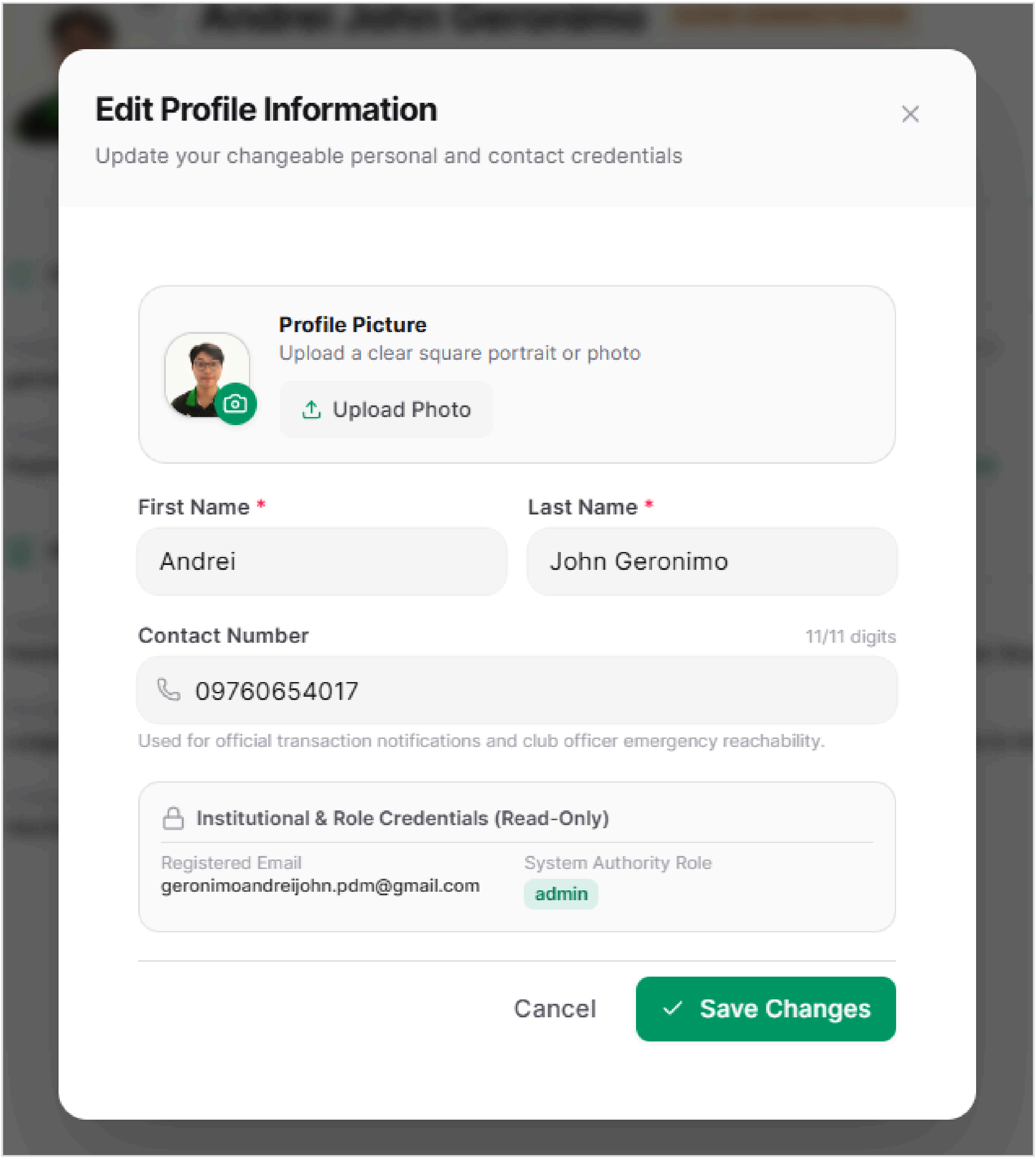
Recognized Student Organization
League of Information Technology Enthusiasts (LITE)

Degree Program
Bachelor of Science in Information Technology (BSIT)

Campus Jurisdiction
Marilao, Bulacan

Figure 32. User Account Profile Page

This figure shows the Account Profile page of the logged-in user. It displays the user's avatar with photo change icon, full name, Super Administrator role badge, email address, verification badges (Account Active & Verified, Google Workspace OAuth), "Edit Profile" and "Security" navigation buttons, Contact & Access Details (registered email, contact number, financial role authority, account status), and Institutional Affiliation details (Pambayang Dalubhasaan ng Marilao, College of Computer Studies, League of Information Technology Enthusiasts, BSIT program, and Marilao, Bulacan jurisdiction). Also here, the user can review their personal account credentials, verify institutional affiliation data, update contact information, and navigate to security settings.



Edit Profile Information

Update your changeable personal and contact credentials

Profile Picture

Upload a clear square portrait or photo

First Name *: Andrei

Last Name *: John Geronimo

Contact Number: 09760654017

11/11 digits

Used for official transaction notifications and club officer emergency reachability.

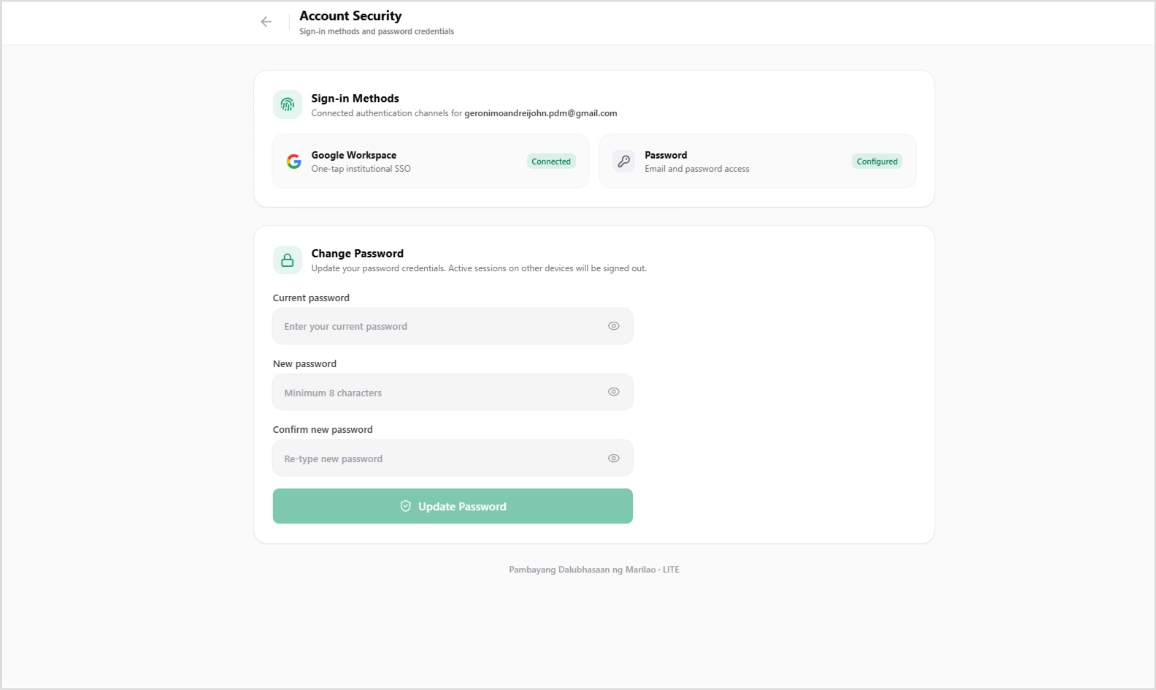
Institutional & Role Credentials (Read-Only)

Registered Email: geronimoandreijohn.pdm@gmail.com

System Authority Role: admin

Figure 33. Edit Profile Information Modal

This figure shows the profile information modal for updating changeable credentials. It contains an avatar upload card with an "Upload Photo" button, input fields for First Name, Last Name, and an 11-digit Contact Number, alongside a read-only Institutional & Role Credentials card showing registered email and system authority role badge, with "Cancel" and "Save Changes" action buttons. Also here, the officer can upload a new profile picture, update their personal contact details, and save modifications to their profile.


Account Security
Sign-in methods and password credentials
Sign-in Methods
Google Workspace
One-tap institutional SSO
Connected
Password
Email and password access
Configured
Change Password
Current password
Enter your current password
New password
Minimum 8 characters
Confirm new password
Re-type new password
Update Password
Pambayang Dalubhasaan ng Marilao · LITE

Connected authentication channels for **geronimoandreijohn.pdm@gmail.com**

Update your password credentials. Active sessions on other devices will be signed out.

Figure 34. Account Security Settings Page

This figure shows the Account Security settings interface. It displays connected Sign-in Methods (Google Workspace One-tap SSO marked Connected, and Password Email/Password access marked Configured), alongside the Change Password card containing input fields for Current password, New password (minimum 8 characters), and Confirm new password with visibility toggles, followed by an "Update Password" button. Also here, the officer can review connected authentication methods, modify their login password, and enforce account security across active sessions.

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